Dissertation : Lévy Stochastic Processes and considerations applied to Finance

Axel Hils - M2 Math4Phys



Abstract

This dissertation investigates the application of advanced **probability theory** to the problem of prediction within complex, jump-diffusion stochastic environments. While the computational aspects of mathematical finance often take precedence in contemporary literature, this research seeks to contribute to scientific progress by re-centering the measure-theoretic foundations that govern information and uncertainty.

Drawing on the fundamental works of **Sato** and **Applebaum**, we analyze the structure of **Lévy processes** through the lens of modern probability. We specifically examine the **Asmussen-Rosiński approximation** as a method for managing the transition between infinite-activity jump measures and their diffusion-based limits. By utilizing **Kallenberg's** rigorous probabilistic framework, we explore how the convergence of these approximations impacts the reliability of predictive models.

Integrating **Fournier's** simulation techniques and **Papapantoleon's** financial applications, this work develops a robust framework for the simulation of **Lévy-driven stochastic differential equations (SDEs)**. The research concludes by formalizing a predictive framework based on the evolution of filtrations and the stability of stochastic integrals. Ultimately, we demonstrate that a deeper commitment to probabilistic rigor not only enhances the accuracy of financial forecasting but also provides a more generalizable contribution to the scientific study of unpredictable systems.

Keywords : Lévy Processes, Probability Theory, Stochastic Prediction, Infinite Divisibility, Distributions, Financial, Approximation, Jumps.

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Introduction

The mathematical modeling of stochastic phenomena has long been dominated by the paradigm of continuity, primarily through the framework of Brownian motion. However, as empirical observations in fields ranging from high-frequency finance to turbulent fluid dynamics have become more refined, the limitations of purely continuous models have become increasingly apparent. The real world is frequently characterized by sudden, discontinuous shocks—events that Brownian motion, with its almost surely continuous paths, is fundamentally unable to capture. This realization has catalyzed a shift toward **Lévy processes**, which serve as a natural and powerful generalization of Brownian motion by allowing for the inclusion of jumps.

A Lévy process is defined by its stationary and independent increments, a property that leads directly to the **Lévy-Khintchine representation**. This representation provides an exhaustive characterization of the process via the Lévy triplet (b, c, ν), where b represents the drift, c the diffusion coefficient, and ν the Lévy measure. The **Lévy-Itô decomposition** further elucidates this structure, partitioning the process into three distinct components: a deterministic linear drift, a continuous Gaussian evolution, and a compensated sum of jumps.

The primary challenge addressed in this dissertation lies in the treatment of **infinite activity** processes. In many modern models, such as the Tempered Stable or Normal Inverse Gaussian families, the Lévy measure ν exhibits a singularity at the origin such that $\nu(\mathbb{R}) = \infty$. This implies that within any finite time interval, the process undergoes an infinite number of infinitesimally small jumps. While theoretically elegant, this property introduces a profound "simulation paradox": a computer, operating in discrete time and with finite resources, cannot explicitly track an infinite sequence of events.

This dissertation investigates the **Asmussen-Rosiński (2001)** approximation as a bridge between this infinite complexity and computational feasibility. We focus on the technique of truncating the Lévy measure at a threshold $\epsilon > 0$ and replacing the discarded "cloud" of small jumps with an equivalent Gaussian component. By matching the variance of this Gaussian term to the truncated portion of the Lévy measure, we aim to preserve the statistical integrity of the process while rendering it simulatable.

The scope of this work extends beyond the processes themselves to **Stochastic Differential Equations (SDEs)** driven by Lévy noise. Building on the convergence frameworks established by **Fournier (2013)**, we analyze how these approximations propagate through dynamical systems. The following chapters will provide :

- A rigorous derivation of the **Lévy-Khintchine formula** and the characteristic exponent :

$$\psi(u) = ibu - \frac{1}{2}cu^2 + \int_{\mathbb{R}\setminus\{0\}} (e^{iux} - 1 - iux\mathbb{1}_{|x|<1})\nu(dx)$$

- A comparative analysis of simple truncation versus **Gaussian substitution** for small-jump management.
- Detailed proofs of convergence for **Euler-Maruyama schemes** in the context of Lévy-driven SDEs.

This dissertation is organized in a progressive manner, moving from the theoretical foundations of Lévy processes to their approximation and applications. We begin by introducing Lévy processes, presenting their definition, fundamental properties, and the Lévy–Khintchine representation, with particular emphasis on infinitely divisible distributions and the role of the Lévy measure. We then develop the Lévy–Itô decomposition, highlighting the distinction between finite and infinite activity processes and analyzing the contribution of small and large jumps. Classical examples, including compound Poisson processes, α-stable Lévy processes, and the variance gamma process, are discussed to illustrate these concepts. The core of the dissertation is devoted to the approximation of small jumps in infinite-activity Lévy processes, where we address the impossibility of exact simulation, introduce truncation techniques, and develop Gaussian approximations. In particular, we provide a detailed study of the results of Asmussen and Rosiński, including assumptions on the Lévy measure, convergence results, and error estimates. Finally, we apply these approximation methods to Lévy-driven stochastic differential equations, presenting both the general framework and numerical schemes, and concluding with convergence results for these approximations.

1 Foundations of Lévy Processes

1.1 Definition and Basic Properties of Lévy Processes

1.1.1 Definition and characterization

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{P})$ be a filtered probability space.

Definition : (Stochastic process)

A family $(X_t)_{t\geqslant 0}$ of random variables is called a stochastic process on $\mathbb{R}$ with parameter $t \in \mathbb{R}_+$.

Remark : $\mathcal{F}_t$ gives information on a process up to time t.

Definition : (Lévy process)

A stochastic process $(X_t)_{t\geqslant 0}$ is called a Lévy process if the following conditions are verified :

1. $\forall n \in \mathbb{N} \quad \forall (t_k)_{k\in[\![0,n]\!]}$ subdivision of $[0,t]$, for $t \geqslant 0$, where $0 = t_0$ and $t_n = t$ with $t_k < t_{k+1} \quad \forall k \in [\![0, n-1]\!]$, $\quad X_{t_{k+1}} - X_{t_k}$ are independent $\forall k \in [\![0, n-1]\!]$.
2. $X_0 = 0 \quad$ a.s.
3. $X_{s+t} - X_s$ is independent of s $\forall s, t \geqslant 0$
4. $\forall\ s \geqslant 0\ \forall\ \varepsilon > 0, \quad \mathbb{P}(|X_s - X_t| > \varepsilon) \xrightarrow[t\to s]{} 0 \quad$ *i.e.* $(X_t)_{t\geqslant 0}$ is stochastically continuous.
5. $\forall\ \omega \in \Omega, \quad X_t(\omega)$ is càdlàg.

Definition : (càdlàg stochastic process)

A càdlàg stochastic process $(X_t)_{t\geqslant 0}$ on a filtered probability space with $X_0 = 0$ is a process verifying :

1. $\forall\ 0 < s < t, \quad X_t - X_s \perp\!\!\!\perp \mathcal{F}_s$
2. $\forall\ t, s \geqslant 0, \quad X_{t+s} - X_t$ is independent of t

One can see that càdlàg means continue à droite linéaire à gauche.
Then comes usual definitions of used objects which are (compound) *Poisson* processes, Martingales, Brownian motions and constant drifts.

Definition : (Random walk)

Consider $(Z_n)_{n\in\mathbb{N}*}$ sequence of independent and identically distributed (*iid*) $\mathbb{R}$-valued random variables. $S_n = \sum_{k=1}^{n} Z_k \quad \forall n \in \mathbb{N}$, namely $S_0 = 0$, is a random walk on $\mathbb{R}$.

1.1.2 First examples and intuition

A classic example of a Lévy process is a Poisson process defined below :

Definition : (Poisson process)

Let $(X_t)_{t\geqslant 0}$ be a stochastic process, $(X_t)_{t\geqslant 0}$ is a Poisson process with parameter λ if it is a Lévy process where $X_t \sim \mathcal{P}(\lambda t) \quad \forall t \geqslant 0$.

In addition to this, the following particular Lévy process can be expressed :

Example : (Lévy "jump-diffusion" process)

Summing a constant drift with a Brownian motion and a compensated compound Poisson process we get for $t \geqslant 0$ the following random variable :

$$L_t = bt + \sqrt{c}W_t + \sum_{k=1}^{N_t}(J_k - \lambda E) \quad \text{with}$$

- $E = \mathbb{E}(J) \quad J = (J_k)_{k\geqslant 1}$ identically and independently distributed sequence of random variables
- $(W_t)_{t\geqslant 0}$ classic Brownian motion
- $N = (N_t)_{t\geqslant 0}$ Poisson process with parameter λ.

Thus process $(L_t)_{t\geqslant 0}$ is called a Lévy jump-diffusion.

1.2 Lévy–Khintchine Formula and Lévy Measure

This formula uses the notion of infinitely divisible distributions/laws which will be defined in the upcoming section. Note that often distributions are viewed in the sense of the analysis tool but here the distributions indeed denote the law of the random variables taken into consideration.
The formula uses the notion of characteristic function of a random variable defined by : $\phi_u(X) = \hat{\mu}(u) = \mathbb{E}[e^{iuX}]$
As a result, the characteristic function can also been seen as the Fourier transform of a distribution.

Consider a Poisson distribution, so for X random variable following $X \sim \mathcal{P}(\lambda)$ we have :

$$\mathbb{P}(X = k) = p(\{k\}) = \frac{\lambda^k e^{-\lambda}}{k!}$$

So,

$$\begin{aligned}\mathbb{E}(e^{iuX}) = \hat{\mu}_{\mathcal{P}}(u) &= \sum_{k=0}^{+\infty} e^{iuk}\mathbb{P}(X = k) = \sum_{k=0}^{+\infty}(e^{iu})^k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= e^{-\lambda}\sum_{k=0}^{+\infty}\frac{(e^{iu}\lambda)^k}{k!} = e^{-\lambda}e^{e^{iu}\lambda} = e^{e^{iu}\lambda-\lambda} = \exp(\lambda(e^{iu}-1))\end{aligned}$$

For a Gaussian distribution *i.e.* $X \sim \mathcal{N}(b, c)$ we have :

$$\mathbb{P}_X(dx) = \frac{e^{\frac{-(x-b)^2}{2c}}}{\sqrt{2\pi c}}dx = \mu(dx)$$

henceforth yields that :

$$\hat{\mu}_{\mathcal{N}}(u) = \int_{\mathbb{R}} e^{iux}\frac{e^{-\frac{(x-b)^2}{2c}}}{\sqrt{2\pi c}}\,dx = \frac{1}{\sqrt{2\pi c}}\int_{-\infty}^{+\infty} e^{iux}e^{-\frac{(x-b)^2}{2c}}\,dx$$

Notice first that if $X \sim \mathcal{N}(b, c)$ then by denoting $X = \sqrt{c}Y + b$ with $Y \sim \mathcal{N}(0, 1)$:

$$\hat{\mu}_{\mathcal{N}}(u) = \mathbb{E}\left(e^{iu(\sqrt{c}Y+b)}\right) = \mathbb{E}\left(e^{iu\sqrt{c}Y}e^{iub}\right) = e^{iub}E\left(e^{iu\sqrt{c}Y}\right)$$

$$\begin{aligned}\mathbb{E}(e^{iu\sqrt{c}Y}) &= \int_{\mathbb{R}} e^{iu\sqrt{c}x}\frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}dx \\ &= \int_{-\infty}^{+\infty}\frac{e^{iu\sqrt{c}x-\frac{x^2}{2}}}{\sqrt{2\pi}}dx = \tilde{\mu}(u)\end{aligned}$$

- For a.e. $x \in \mathbb{R}, u \mapsto e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}$ is C^1 on $\mathbb{R}$
- $\forall u \in \mathbb{R}, x \mapsto e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}$ is piecewise continuous and integrable over $\mathbb{R}$ (as $|e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}| \leqslant e^{-\frac{x^2}{2}}$ which is simply the standard Gaussian integral when integrated over $\mathbb{R}$).
  $x \mapsto ix\sqrt{c}\ e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}$ is piecewise continuous over $\mathbb{R}$.
- $\forall u$, for a.e. $x \in \mathbb{R}, |ix\sqrt{c}\ e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}| \leqslant |x|\sqrt{c}\ e^{-\frac{x^2}{2}}$ and $x \mapsto x\sqrt{c}\ e^{-\frac{x^2}{2}}$ is also integrable over $\mathbb{R}$ by performing integration by parts and computing standard Gaussian integral.

Finally, by regularity of parameterized integral theorem :

$$\tilde{\mu}'(u) = \frac{1}{\sqrt{2\pi}}\int_0^{+\infty} ix\sqrt{c}\ e^{-\frac{x^2}{2}}e^{iux}dx = \frac{i\sqrt{c}}{\sqrt{2\pi}}\int_0^{+\infty} xe^{-\frac{x^2}{2}}e^{iux}dx$$

so by integration by parts :

$$\begin{aligned}
\tilde{\mu}'(u) &= \frac{i\sqrt{c}}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{iu\sqrt{c}x}\ xe^{-\frac{x^2}{2}}dx \\
&= \frac{i\sqrt{c}}{\sqrt{2\pi}}\left(\cancel{\left[-e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}\right]_{-\infty}^{+\infty}} - \int_{-\infty}^{+\infty} -iu\sqrt{c}\ e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}dx\right) \\
&= \frac{i\sqrt{c}}{\sqrt{2\pi}}\ iu\sqrt{c}\int_{-\infty}^{+\infty} e^{iu\sqrt{c}x}e^{-\frac{x^2}{2}}dx \\
&= -cu\tilde{\mu}(u)
\end{aligned}$$

Hence, $\tilde{\mu}$ solve the first order linear differential equation :

$$\tilde{\mu}'(u) + cu\tilde{\mu}(u) = 0$$

Therefore :

$$\tilde{\mu}(u) = \tilde{\mu}(0)e^{-\frac{cu^2}{2}}$$

And $\tilde{\mu}(0) = \mathbb{E}(1) = 1$ so :

$$\hat{\mu}_{\mathcal{N}}(u) = e^{iub}\tilde{\mu}(u) = e^{iub}e^{-c\frac{u^2}{2}} = e^{iub}e^{-c\frac{u^2}{2}}$$

And, for a compound Poisson process $(X_t)_{t\geqslant 0}$, one has that :

$$\hat{\mu}_{\mathcal{P}}(u) = \mathbb{E}(e^{iuX_t}) = \exp(\lambda(\sigma(u) - 1))$$

where σ is a finite distribution $\sigma(\{0\}) = 0$
So we can see that :

$$\widehat{\mu_{\mathcal{P}} * \mu_{\mathcal{N}}}(u) = \hat{\mu}_{\mathcal{P}}\hat{\mu}_{\mathcal{N}}(u) = \exp\left(iub - \frac{u^2c}{2} + \int_{\mathbb{R}}(e^{iux} - 1)\,\nu(dx)\right)$$

where ν is a finite measure and $\nu(\{0\}) = 0$.

1.2.1 Infinitely divisible distributions

The definition is given by :

Definition : (Infinitely divisible distributions)

The law $\mathbb{P}_X$ (or distribution μ) of a random variable X is infinitely divisible if $\exists\ n$ identically and independently distributed random variables $X_i^{1/n}$ such that :

$$X = \sum_{i=1}^{n} X_i^{1/n}$$

An equivalent definition of this phenomenon is the law P_X being written :

$$\mathbb{P}_X = \underbrace{\mathbb{P}_{X^{1/n}} * \cdots * \mathbb{P}_{X^{1/n}}}_{n \text{ times}}$$

where $P_{X^{1/n}}$ is the law of another random variable $X^{1/n}$ and $*$ denotes the convolution product.

Typically for two measures λ and ρ we have for $A \in \mathcal{B}(\mathbb{R})$:

$$(\lambda * \rho)(A) = \int_{\Omega} \lambda(A - y)\rho(dy)$$

An example can be provided to image this definition :

Example : (Lévy process distribution)

For a Lévy process $(X_t)_{t\geqslant 0}$, X_t has an infinitely divisible distribution. Indeed, denote $t_k = \frac{kt}{n}$, so $(t_k)_{k\in[\![0,n]\!]}$ is a subdivision of $[0,t]$. By the property of independent increments, we have:

$$X_t = \sum_{k=1}^{n} X_{t_k} - X_{t_{k-1}} = (X_{t_n} - X_{t_{n-1}}) + (X_{t_{n-1}} - X_{t_{n-2}}) + \cdots + (X_{t_1} - X_{t_0})$$

where $X_{t_n} = X_t$ and $X_{t_0} = 0 = X_0$. So $(X_{t_k} - X_{t_{k-1}})_{k\in[\![1,n]\!]}$ is a sequence of identically and independently distributed random variables.

1.2.2 Lévy–Khintchine representation

The two following results are regarded as Theorem 4.7 and Lemma 8.6 in the book of Ken-Iti Sato.

Theorem : (Representation of characteristic functions of infinitely divisible distributions)

The law $\mathbb{P}_X$ of a random variable X is infinitely divisible if and only if there exists a unique triplet (b, c, ν), with $b \in \mathbb{R}$, $c \in \mathbb{R}_{\geqslant 0}$ and a measure satisfying $\nu(\{0\}) = 0$ and $\int_{\mathbb{R}}(1 \wedge |x|^2)\nu(dx) < \infty$, such that:

$$\mathbb{E}[e^{iuX}] = \exp\left[ibu - \frac{u^2 c}{2} + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx)\right] \tag{1}$$

(1) is called the *Lévy–Khintchine* representation of the characteristic function of X with generating triplet (b, c, ν).

Remark : There is a one-to-one correspondance between infinitely divisible distributions and generating triplets. To prove this, first recall Taylor-Laplace formula for any class $\mathcal{C}^{n+1}$ map :

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + \int_a^b \frac{(b-t)^n}{n!} f^{(n+1)}(t)dt \quad (*)$$

From Sato's book we have the following lemma :

Lemma

$\forall\ u \in \mathbb{R},\ n \in \mathbb{N}$

$$e^{iu} = \sum_{k=0}^{n-1} \frac{(iu)^k}{k!} + \theta \frac{|u|^n}{n!} \quad \text{for } \theta \in \mathbb{C},\ |\theta| \leqslant 1$$

Proof : For $(*)$ applied to exp, we get: $a = 0,\ b = u$

$$e^{iu} = \sum_{k=0}^{n-1} \frac{(iu)^k}{k!} + \int_0^u \frac{(u-v)^{n-1}}{(n-1)!} i^n e^{iv} dv = \sum_{k=0}^{n-1} \frac{(iu)^k}{k!} + \frac{i^n}{(n-2)!} \int_0^u (u-v)^{n-1} e^{iv} dv$$

□

This result is used in the following proof of the *Lévy-Khintchine* formula studied in one real dimension *i.e.* on $\mathbb{R}$.

Proof (the triplet is unique) : Suppose that X has its characteristic function expressed as in (1) with b, c and ν. Notice that :

$$\mathbb{E}(e^{iuX}) = \exp\left(ibu - \frac{cu^2}{2} + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx) \right) \text{ then we have}$$

If $|x| < 1$: (by previous lemma for $n = 2$)

$$\begin{aligned} |e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}| = |e^{iux} - 1 - iux| &= |1 + iux + \theta \frac{|u|^2|x|^2}{2} - iux - 1| \\ &= |\theta \frac{|u|^2|x|^2}{2}| \leqslant \frac{|u|^2|x|^2}{2} = \frac{|u|^2|x|^2}{2} \mathbb{1}_{\{|x|<1\}} \end{aligned}$$

If $|x| \geqslant 1$:

$$|e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}| = |e^{iux} - 1| \leqslant |e^{iux}| + 1 \leqslant 1 + 1 = 2\ \mathbb{1}_{\{|x|\geqslant 1\}}$$

As $\frac{|u|^2|x|^2}{2} \mathbb{1}_{\{|x|<1\}}$ and $2\ \mathbb{1}_{\{|x|\geqslant 1\}}$ are positive quantities, then: $\forall\ x \in \mathbb{R}$

$$|e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}| \leqslant \frac{1}{2}|u|^2|x|^2 \mathbb{1}_{\{|x|<1\}}(x) + 2\ \mathbb{1}_{\{|x|\geqslant 1\}}(x) \tag{2}$$

Then by Lebesgue convergence theorem, expression defined in (2) is continuous in u. So then :

$$\begin{aligned} \ln(\mathbb{E}(e^{iusX})) &= -\frac{1}{2}s^2u^2c + isub + \int_{\mathbb{R}} e^{iux} - 1 - iux\ \mathbb{1}_{\{|x|<1\}}(x)\nu(dx) \quad \text{for } s \in \mathbb{R} \\ &\leqslant -\frac{1}{2}s^2u^2c + isub + \int_{\mathbb{R}} \frac{1}{2}|u|^2|x|^2 \mathbb{1}_{\{|x|<1\}}(x) + 2 \cdot \mathbb{1}_{\{|x|\geqslant 1\}}(x)\nu(dx) \end{aligned}$$

So, by Lebesgue theorem again :

$$\frac{1}{s^2} \ln(\mathbb{E}(e^{iusX})) \xrightarrow[s\to+\infty]{} -\frac{1}{2}u^2c$$

Henceforth, c is uniquely defined by $\mathbb{E}(e^{iuX})$, *i.e.*, the distribution of X. Set now : $\psi(u) = \ln(\mathbb{E}(e^{iuX})) + \frac{1}{2}u^2c$. For $C = [-1, 1]$, one can prove that :

$$\int_C (\psi(u) - \psi(u+w))dw = 2\int_{\mathbb{R}} e^{iux} \left(1 - \frac{\sin(x)}{x}\right) \nu(dx) \tag{3}$$

and

$$\frac{\sin(x)}{x} = \begin{cases} \frac{\sin(x)}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$\begin{aligned}\psi(u)-\psi(u+w) &= \ln(\mathbb{E}(e^{iuX}))+\frac{1}{2}u^2c-\ln(\mathbb{E}(e^{i(u+w)X}))+\frac{1}{2}(u+w)^2c\\ &= iub+\int_{\mathbb{R}}(e^{iux}-1-iux\mathbb{1}_{\{|x|<1\}}(x))\nu(dx)-i(u+w)b-\int_{\mathbb{R}}(e^{i(u+w)x}-1-i(u+w)x\mathbb{1}_{\{|x|<1\}}(x))\nu(dx)\\ &= \int_{\mathbb{R}}(e^{iux}-e^{i(u+w)x}+iwx\mathbb{1}_{\{|x|<1\}}(x))\nu(dx)-iwb\end{aligned}$$

Noticing that :

$$\begin{aligned}|e^{iux}-e^{i(u+w)x}+iwx\mathbb{1}_{\{|x|<1\}}(x)| &\leqslant |1+e^{iwx}-1-e^{i(u+w)x}+iwx|\\ &\leqslant |1-e^{iwx}+iwx|+|wx||1-e^{iux}|\\ &\leqslant \frac{1}{2}|w|^2|x|^2+|w||x|^2|u|\end{aligned}$$

Then, Fubini's theorem gives :

$$\int_C(\psi(u)-\psi(u+w))dw=\int_{\mathbb{R}}e^{iux}\nu(dx)\int_C(1-e^{iwx})dw$$

Which proves exactly formula (3). As one can notice that :

$$\begin{aligned}\int_C(1-e^{iwx})dw=\int_{-1}^{1}(1-e^{iwx})dw &= \int_{-1}^{1}1dw-\int_{-1}^{1}e^{iwx}dw\\ &= 2-\left[\frac{e^{iwx}}{ix}\right]_{-1}^{1}\\ &= 2-\left(\frac{e^{ix}-e^{-ix}}{ix}\right)=2-\frac{2}{x}\frac{e^{ix}-e^{-ix}}{2i}\\ &= 2\left(1-\frac{\sin(x)}{x}\right)\end{aligned}$$

by Euler formula for complex exponential. By setting :

$$\zeta(dx)=2\left(1-\frac{\sin(x)}{x}\right)\nu(dx)$$

ζ is a finite measure as :

$$\begin{aligned}\frac{\sin(x)}{x} &= \left(x-\frac{x^3}{6}+\underset{x\to 0}{o}(x^3)\right)\times\frac{1}{x}=1-\frac{x^2}{6}+\underset{x\to 0}{o}(x^2)\\ &= 1-\frac{x^2}{6}+\underset{x\to 0}{O}(x^4)\end{aligned}$$

In the case of distributions, one can also see that :

$$\int_{\mathbb{R}}e^{iux}2\left(1-\frac{\sin(x)}{x}\right)\nu(dx)=\int_{\mathbb{R}}e^{iux}\int_C(1-e^{iwx})dw=\int_{\mathbb{R}}e^{iux}\zeta(x)dx$$

is simply the Fourier transform of probability measure ζ. Then by isomorphism property of Fourier transformation map : $\mathcal{F}\colon \mu\longmapsto\mathcal{F}(\mu)$, ζ is uniquely defined by P_X, so ν is also uniquely determined by P_X. And b is determined from P_X and c. Hence uniqueness of generating triplet (b,c,ν).

□

In the *Lévy–Khintchine* formula, the expression of characteristic function can be read as well :

$$\mathbb{E}(e^{iuX}) = \exp\left(iub_\chi - \frac{1}{2}u^2c + \int_{\mathbb{R}}(e^{iux} - 1 - iux\chi(x))\nu(dx)\right) \tag{4}$$

where :

$$b_\chi = b + \int_{\mathbb{R}} x(\chi(x) - \mathbb{1}_D(x))\nu(dx) \quad \text{where } \chi \text{ is a bounded measurable}$$

$$\mathbb{R}\text{-valued function.}$$

Proof :

$\boxed{\Longrightarrow}$ Suppose that X is an infinitely divisible random variable with associated distribution μ. Then, by setting :

$$\widehat{\mu}_n(u) = \exp\left(iub - \frac{1}{2}u^2c + \int_{|x|>\frac{1}{n}}(e^{iux} - 1 - iux\mathbb{1}_D(x))\nu(dx)\right)$$

By previous studies, $\widehat{\mu}_n$ is the characteristic function of an infinitely divisible distribution as it is seen as the convolution of a compound Poisson and a Gaussian distribution.

$$\widehat{\mu}_n(u) \xrightarrow[n\to\infty]{} \exp\left(iub - \frac{1}{2}u^2c + \int_{\mathbb{R}}(e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx)\right) = \widehat{\mu}(u)$$

Hence, by previous step $\widehat{\mu}$ is continuous so as the limit of infinite divisible distributions, $\widehat{\mu}$ is the characteristic function with *Lévy–Khintchine* representation of an infinitely divisible distribution.

□

Theorem : (Convergence of Infinitely Divisible Distributions)

Let $c(x)$ be a bounded, continuous, $\mathbb{R}$-valued function satisfying :

- $c(x) = 1 + o(|x|)$ as $|x| \to 0$
- $c(x) = O\left(\frac{1}{|x|}\right)$ as $|x| \to \infty$

Suppose $(\mu_n)_{n\in\mathbb{N}*}$ is a sequence of infinitely divisible distributions with *Lévy–Khintchine* triplets (b_n, c_n, ν_n). Then, μ_n converges weakly to μ ($\mu_n \to \mu$) **if and only if** μ is infinitely divisible with triplet (b, c, ν) and the following conditions hold :

1. **Lévy Measure Convergence :** $\forall f \in C_\#$ (bounded continuous functions where $f(x) \xrightarrow[|x|\to 0]{} 0$) :

$$\int_{\mathbb{R}} f(x)\nu_n(dx) \xrightarrow[n\to\infty]{} \int_{\mathbb{R}} f(x)\nu(dx)$$

2. **Diffusion Convergence :**

$$c_n \xrightarrow[n\to\infty]{} c$$

3. **Drift Convergence :**

$$b_n \xrightarrow[n\to\infty]{} b$$

Note that condition (2) becomes more complex for $\mathbb{R}^d$ distributions as scalar product is introduced and diffusion coefficient becomes a matrix.

Proof :

$\boxed{\Longleftarrow}$ Suppose that μ is an infinitely divisible distribution. Introduce $(t_n)_n \subset \mathbb{R}^+$ where $\lim_{n\to\infty} t_n = 0$. Define

the following characteristic functions of probability measure sequence $(\mu_n)_n$:

$$\widehat{\mu}_n(u) = \exp\left(\frac{1}{t_n}((\widehat{\mu}(u))^{t_n} - 1)\right) = \exp\left(\frac{1}{t_n}\int_{\mathbb{R}}(e^{iux} - 1)\mu^{t_n}(dx)\right)$$

One can notice that μ_n is compound Poisson. Moreover :

$$\widehat{\mu}_n(u) = \exp\left(\frac{1}{t_n}(e^{t_n \ln(\widehat{\mu}(u))} - 1)\right) = \exp\left(\frac{1}{t_n}(t_n \ln(\widehat{\mu}(u)) + O(t_n^2))\right)$$

So $\forall\ u \in \mathbb{R}$,

$$\widehat{\mu}_n(u) \xrightarrow[n\to\infty]{} \widehat{\mu}(u)$$

as μ_n has representation (4) by the previous Theorem, $\widehat{\mu}$ has *Lévy–Khintchine* representation with generating triplet (b, c, ν). Therefore being written in the form of (1).

□

1.2.3 Lévy triplet

The Lévy triplet is usually the name given to the previous generating triplet (b, c, ν) defined in *Lévy–Khintchine* representation where we have $b \in \mathbb{R}$ the drift term, $c \in \mathbb{R}_+$ the diffusion coefficient and μ the Lévy measure. Here follows a detailed example for one reference law :

Example : (Gamma distribution)

For a Gamma distribution, density reads : $f(x) = \frac{x^{k-1}e^{-\frac{x}{\theta}}}{\Gamma(k)\theta^k}\mathbb{1}_{\{x\geqslant 0\}}$
$\forall k, \theta > 0$, so for $X \sim \Gamma(k, \theta)$:
Consider : $\varphi(z) = \int_0^{+\infty} e^{-\frac{x}{\theta}}e^{zx}x^{k-1}dx = \int_0^{+\infty} e^{-x(\frac{1}{\theta}-z)}x^{k-1}dx$

On $C = \{z \in \mathbb{C} \mid \mathrm{Re}(z) < \frac{1}{\theta}\}$, f is well-defined and analytic. Moreover notice that for $z \in \mathbb{R}$ where $z < \frac{1}{\theta}$ then $f(z)$ exists and gives, by w-substitution: $w = x(\frac{1}{\theta} - z)$ $\begin{array}{c|c} x & 0 \to +\infty \\ \hline w & 0 \to +\infty \end{array}$ $\frac{dx}{dw} = \frac{1}{\frac{1}{\theta}-z}$

$$\begin{aligned}\varphi(z) &= \int_0^{+\infty}\left(\frac{w}{\frac{1}{\theta}-z}\right)^{k-1}e^{-w}\frac{dw}{\frac{1}{\theta}-z} = \frac{1}{(\frac{1}{\theta}-z)^k}\int_0^{+\infty}w^{k-1}e^{-w}dw \\ &= \frac{\Gamma(k)}{(\frac{1}{\theta}-z)^k}\end{aligned}$$

As $C' = \{z \in \mathbb{R} \mid z < \frac{1}{\theta}\} \subset C$ and is obviously a connected subset with an accumulation point (In fact this part of the real line has infinitely many limit points), so Identity theorem here yields that over $C = \{z \in \mathbb{C} \mid \mathrm{Re}(z) < \frac{1}{\theta}\}$, $\varphi(z) = \int_0^{+\infty} x^{k-1}e^{-x(\frac{1}{\theta}-z)}dx = \frac{\Gamma(k)}{(\frac{1}{\theta}-z)^k}$

For $z = iu$, $\mathrm{Re}(z) = 0 < \frac{1}{\theta}$. Thus $\varphi(iu) = \int_0^{+\infty} x^{k-1}e^{-x(\frac{1}{\theta}-iu)}dx = \frac{\Gamma(k)}{(\frac{1}{\theta}-iu)^k}$

Henceforth : $\hat{\mu}(u) = \frac{1}{\Gamma(k)\theta^k}\varphi(iu) = \frac{1}{\Gamma(k)\theta^k}\frac{\Gamma(k)}{(\frac{1}{\theta}-iu)^k} = \frac{\theta^k}{\theta^k} \times \frac{1}{(1-i\theta u)^k} = (1 - i\theta u)^{-k}$
Introduce Laplace transform of μ :

$$L_\mu(v) = \int_0^{+\infty} e^{-vx}\mu(dx)$$

So by expression of $\mu = \mathbb{P}_X$ for $X \sim \Gamma(k, \theta)$:

$$\begin{aligned}L_\mu(v) &= \int_0^{+\infty} e^{-vx}\frac{e^{-\frac{x}{\theta}}x^{k-1}}{\Gamma(k)\theta^k}\mathbb{1}_{\mathbb{R}_+}(x)dx \\ &= \frac{1}{\Gamma(k)\theta^k}\int_0^{+\infty} x^{k-1}e^{-x(\frac{1}{\theta}+v)}dx\end{aligned}$$

Example : (Gamma distribution)

By change of variable set : $z = x\left(\frac{1}{\theta} + v\right)$
so $\frac{dz}{dx} = \frac{1}{\theta} + v$

x	0	$+\infty$
z	0	$+\infty$

Hence :

$$\begin{aligned} L_\mu(v) &= \frac{1}{\Gamma(k)\theta^k}\int_0^{+\infty}\left(\frac{z}{\frac{1}{\theta}+v}\right)^{k-1} e^{-z}\frac{dz}{\frac{1}{\theta}+v} \\ &= \frac{\theta^k}{\Gamma(k)\theta^k} \times \frac{1}{(1+\theta v)^k}\int_0^{+\infty} z^{k-1}e^{-z}dz \\ &= \frac{\Gamma(k)}{\Gamma(k)}\frac{1}{(1+\theta v)^k} = (1+\theta v)^{-k} \end{aligned}$$

$\ln((1+\theta v)^{-k}) = -k\ln(1+\theta v)$

So as :

$$\begin{aligned} \ln(1+\theta v) &= \int_0^v \frac{1}{\frac{1}{\theta}+y}dy = \int_0^v\int_0^{+\infty} e^{-x\left(\frac{1}{\theta}+y\right)}dxdy \\ &= \int_0^{+\infty} e^{-\frac{x}{\theta}}\int_0^v e^{-xy}dydx = \int_0^{+\infty} e^{-\frac{x}{\theta}}\left[\frac{e^{-xy}}{-x}\right]_0^v dx \\ &= -\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}}{x}(e^{-vx}-1)dx \end{aligned}$$

Whence :

$$\begin{aligned} L_\mu(v) &= \exp(\ln((1+\theta v)^{-k})) = \exp(-k\ln(1+\theta v)) \\ &= \exp\left(k\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}}{x}(e^{-vx}-1)dx\right) \end{aligned}$$

By extension to $\{Re(\omega) \leqslant 0 \mid \omega \in \mathbb{C}\}$ on the complex plane :

$$L_\mu(-\omega) = \int_0^{+\infty} e^{\omega x}\mu(dx) = \exp\left(k\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}}{x}(e^{\omega x}-1)dx\right)$$

By setting $\omega = iu$, the previous yields :

$$\mathbb{E}(e^{iuX}) = \hat{\mu}(u) = \exp\left(k\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}}{x}(e^{iux}-1)dx\right)$$

So one notices that this characteristic function is represented in *Lévy–Khintchine* form with generating triplet (b, c, ν) given by :

- $b = 0$
- $c = 0$
- $\nu(dx) = \frac{ke^{-\frac{x}{\theta}}}{x}\mathbb{1}_{\mathbb{R}_+}dx$

So for a Lévy process one gets that :

Theorem :

For a Lévy process $(L_t)_{t\geqslant 0}$ the characteristic function reads :

$$\mathbb{E}(e^{iuL_t}) = \exp\left(t\left(iub - \frac{c}{2}u^2 + \int_{\mathbb{R}}(e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx)\right)\right) = \exp(t\psi(u)) \tag{5}$$

$\psi(u)$ is the characteristic exponent of an infinitely divisible random variable L_1.

Proof : Introduce $\phi_t(u) = \phi_u(t) = \mathbb{E}(e^{iuL_t})$, one can notice that:

$$\phi_u(t+s) = \mathbb{E}(e^{iuL_{t+s}}) = \mathbb{E}(e^{iu(L_{t+s}-L_s)}e^{iuL_s}) = \mathbb{E}(e^{iu(L_{t+s}-L_s)})\mathbb{E}(e^{iuL_s})$$

by independence of $L_{t+s} - L_s$ and L_s, furthermore as $L_{t+s} - L_s$ is independent of s, we have: $\mathbb{E}(e^{iu(L_{t+s}-L_s)}) = \mathbb{E}(e^{iuL_t})$ so: $\phi_u(t+s) = \phi_u(t)\phi_u(s)$ $(*)$ and the only solution to $(*)$ is: $\phi_u(t) = e^{t\psi(u)}$ where $\psi : \mathbb{R} \to \mathbb{C}$ continuous. Applying this for $t = 1$ one gets: $\phi_u(1) = e^{\psi(u)} = \mathbb{E}(e^{iuL_1})$, as L_1 is an infinitely divisible random variable by Example in section dealing with infinitely divisible distributions. So by *Lévy-Khintchine* representation theorem : $\psi(u)$ is the characteristic exponent of L_1 of the form: $\psi(u) = iub - \frac{1}{2}u^2c + \int_{\mathbb{R}}(e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx)$

□

1.2.4 Interpretation of the Lévy measure

For $A \in \mathcal{B}(\mathbb{R}), A \not\ni \{0\}$, introduce the random measure of the jumps of a Lévy process L: $\mu^L(\omega; t, A) = \mu^L(t, A)$

$$\mu^L(\omega; t, A) = \sum_{s\leqslant t}\mathbb{1}_A(\Delta L_s(\omega)) \quad \text{where we define} \quad \Delta L_s = L_s - L_{s^-}, \quad L_{s^-} = \lim_{v\to s, v<s} L_v$$
$$= \text{card}\left(\{0 \leqslant s \leqslant t \mid \Delta L_s(\omega) \in A\}\right) \quad \Delta L = (\Delta L_t)_{t\geqslant 0} \text{ is the jump process associated to } L = (L_t)_{t\geqslant 0}$$

Definition : (Lévy measure)

The measure defined by : $\nu(A) = \mathbb{E}(\mu^L(1, A))$ is the Lévy measure associated to a Lévy process $(L_t)_{t\geqslant 0}$.

By definition of the process manipulated objects, the Lévy measure of a Lévy process can be seen as the expectation of number of jumps of certain heights among a Borelian set of real numbers *i.e.* the expectation of number of jumps that can have specific heights by choosing or fixing a set of heights (A) under time $t = 1$ (or in a time interval of length 1).

2 Lévy–Itô Decomposition

2.1 Decomposition theorem

Note that previously, we assumed presence of filtration on a probability space, this is not mandatory here to have an understanding of the latter definitions. Before stating the decomposition result, a few objects need to be defined in order to understand the theorem.

Definition : (Compound Poisson process)

For $(S_n)_{n\geqslant 0}$ random walk and $(N_t)_{t\geqslant 0}$ a Poisson process, then $(X_t = S_{N_t})_{t\geqslant 0}$ is a compound Poisson process. Notice that $(S_n)_{n\geqslant 0}$ and $(N_t)_{t\geqslant 0}$ are independent.

In the sense of a previous example, one has the following :

Definition : (Compensated compound Poisson process)

For $(S_n)_{n\geqslant 0}$ random walk with associated $Y = (Y_i)_{i\in\mathbb{N}}$ i.i.d. sequence of random variables, and $(N_t)_{t\geqslant 0}$ a Poisson process, then $(X_t = \sum_{i=1}^{N_t}(Y_i - \lambda\mathbb{E}(Y))_{t\geqslant 0}$ is a compensated compound Poisson process. Notice that $(S_n)_{n\geqslant 0}$ and $(N_t)_{t\geqslant 0}$ remain independent.

Remark : The reason why we compensate a compound Poisson is to cancel the deterministic drift of such a process. Indeed, removing the expectation value of the process creates a new process with, thus, mean 0 of any time t, therefore making it a Martingale. To detail a bit more on the deterministic aspect, when we are seeking to separate parts of a Lévy process in the *Lévy-Itô* decomposition, we wish to isolate the random part hence the reason why we compensate small jumps. And finally, another pro of such an operation is for approximation of small jumps, namely this helps center the small jump component to a mean 0 process contributing it to look like a Brownian motion.

Definition : (Brownian motion)

A stochastic process $(B_t)_{t\geqslant 0}$ is called a Brownian motion if it satisfies :

- $\forall t > 0 \quad t \mapsto B_t(\omega)$ is continuous, $B_0 = 0$
- $\forall\ 0 \leqslant s < t \quad B_t - B_s$ follows a Gaussian law of parameters $(0, t-s)$.
- $\forall n \in \mathbb{N} \quad \forall (t_k)_{k\in 0,n}$ subdivision of $[0,t]$ where $0 = t_0$ and $t_n = t$ with $t_k < t_{k+1} \quad \forall k \in [\![0, n-1]\!], \quad X_{t_{k+1}} - X_{t_k}$ are independent $\forall k \in [\![0, n-1]\!]$.

Definition : (Martingale)

A stochastic process $(X_t)_{t\geqslant 0}$ is said to be a martingale if it verifies :

- X_t is measurable with respect to $\mathcal{F}_t \quad \forall\ t \geqslant 0$
- $\mathbb{E}(|X_t|) < +\infty \quad \forall\ t \geqslant 0$
- $\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s \quad \forall\ 0 \leqslant s < t.$

$(X_t)_{t\geqslant 0}$ is said to be an $(\mathcal{F}_t)$-martingale. Furthermore, if $\mathbb{E}(|X_t|^2) < +\infty$, $(X_t)_{t\geqslant 0}$ is a square-integrable $(\mathcal{F}_t)_{t\geqslant 0}$-martingale.

Remark : As a Brownian motion and a compensated compound Poisson process are martingales, a Lévy jump-diffusion becomes a martingale if and only if $b = 0$.
Also, the Lévy measure has no mass at the origin but singularities occur around the origin *i.e.* infinitely many small jumps whereas the mass is bounded away from the origin yielding only finitely many large jumps.

Whence, the decomposition theorem yields :

Theorem : (*Lévy-Itô* decomposition)

For a triplet (b, c, ν), $b \in \mathbb{R}$, $c \geqslant 0$, ν finite measure such that :

- $\nu(\{0\}) = 0$
- $\int_{\mathbb{R}}(1 \wedge |x|^2)\nu(dx) < +\infty$

Then, $\exists(\Omega, \mathcal{F}, P)$ probability space on which 4 independent Lévy processes $L^{(1)}, L^{(2)}, L^{(3)}, L^{(4)}$ verifying :

- $L^{(1)}$ is a constant drift
- $L^{(2)}$ is a Brownian motion
- $L^{(3)}$ is a compound Poisson process
- $L^{(4)}$ is (square-integrable) martingale (pure jump) with a.s. countable number of jumps of magnitude less than 1 on every finite time interval.

And $L = L^{(1)} + L^{(2)} + L^{(3)} + L^{(4)}$ is a Lévy process with Lévy exponent :

$$\psi(u) = \exp\left(iub - \frac{1}{2}u^2c + \int_{\mathbb{R}}(e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}})\nu(dx)\right) \quad \forall u \in \mathbb{R}.$$

Therefore, the Lévy exponent can be decomposed into 4 parts given by: $\psi = \psi_1 + \psi_2 + \psi_3 + \psi_4$ where :

$$\psi_1(u) = iub \qquad \psi_3(u) = \int_{|x|\geqslant 1}(e^{iux} - 1)\nu(dx)$$
$$\psi_2(u) = -\frac{1}{2}u^2c \qquad \psi_4(u) = \int_{|x|<1}(e^{iux} - 1 - iux)\nu(dx)$$

If one uses the jump associated random measure then one gets that, for a Lévy process $(L_t)_{t\geqslant 0}$:

$$L_t = bt + \sqrt{c}W_t + \int_0^t\int_{|x|\geqslant 1} x\,\mu^L(ds, dx) + \int_0^t\int_{|x|<1} x(\mu^L - \nu^L)(ds, dx)$$

furthermore, $\nu^L(ds, dx) = \nu(dx)ds$ and L gives out 4 independent Lévy processes listed in *Lévy-Itô* decomposition.
Regarding the measure counting jumps of a Lévy process, μ^L is a Poisson random measure with intensity $\nu(A) = \mathbb{E}(\mu^L(1, A))$ for Poisson process $(\mu^L(t, A))_{t\geqslant 0}$.

2.2 Finite and infinite activity

The activity of a Lévy process depends on all type of jumps of the process here on small and big jumps. Leading to diverse results dealing with the latter. An elegant way of evaluating type of activity of a process is given by studying the behavior of the Lévy measure :

Proposition: (Activity type)

Let L be a Lévy process with associated triplet (b, c, ν) :

1. $\nu(\mathbb{R}) < +\infty \implies L$ is of **finite activity**. (Almost all paths of L have a finite number of jumps on every compact interval.)
2. $\nu(\mathbb{R}) = +\infty \implies L$ is of **infinite activity**. (Almost all paths of L have an infinite number of jumps on every compact interval.)

The activity can thus be cut into two sections: small and large jumps. For small jumps, the following arises : (seeing that the Lévy measure is not the only parameter taken into account)

Proposition: (Variation type)

Consider a Lévy process L with associated triplet (b, c, ν). Then :

1. If $c = 0$ and $\int_{|x|\leqslant 1} |x|\, \nu(dx) < +\infty$ then almost all paths of L have **finite variation**.
2. If $c \neq 0$ and $\int_{|x|\leqslant 1} |x|\, \nu(dx) = +\infty$ then almost all paths of L have **infinite variation**.

Furthermore, the Lévy measure gives information about the moments of a Lévy process.
So the following proposition can come in handy :

Proposition : (Number of jumps)

Consider a Lévy process $L = (L_t)_{t\geqslant 0}$ with generating triplet (b, c, ν). We have the following assertions :

1. $\mathbb{E}(|L_t|^p) < +\infty \iff \int_{|x|\geqslant 1} |x|^p \nu(dx) < +\infty$ (p-th moment)
2. $\mathbb{E}(e^{pL_t}) < +\infty \iff \int_{|x|\geqslant 1} e^{px} \nu(dx) < +\infty$ (p-th exponential moment)

for any $p \geqslant 0$.

Remark : One can notice that big jumps refer to jumps absolutely ($|\cdot|$) larger than 1.

2.3 Small versus large jumps

When a Lévy process has a finite number of jumps, it is said to have **finite activity**, as a result, if it has an infinite number of jumps we speak of **infinite activity** Lévy processes.
So for finite variation jumps, *i.e.* small jumps, this is verified if and only if :

$$\int_{|x|<1} |x| \nu(dx) < +\infty$$

So for this case the Lévy-Itô decomposition is given by :

$$L_t = \gamma t + \sqrt{c} W_t + \int_0^t \int_{\mathbb{R}} x \mu^L(ds, dx) \quad \text{where} \quad \gamma = b - \int_0^1 x \nu(dx)$$

3 Classical Examples of Lévy Processes

3.1 Compound Poisson processes

By definition in the previous section of a compound Poisson process, conditions 2., 4. and 5. of axioms verified by Lévy processes are shown so the following proposition can be written :

Proposition

A compound Poisson process $(X_t)_{t\geqslant 0}$ as defined previously is a Lévy process.

Proof : We need to show veracity of conditions 1. and 3. of the definition of a Lévy process in the first section. Let $B_0, \dots, B_k \in \mathcal{B}(\mathbb{R})$ where $k \in \mathbb{N}$ and $B \in \mathcal{B}(\mathbb{R})$, notice that for $t \geqslant 0$:

$$\mathbb{P}(X_t \in B) = \mathbb{P}(S_{N_t} \in B) = \sum_{n=0}^{+\infty} \mathbb{P}([N_t = n] \cap [S_n \in B]) = \sum_{n=0}^{+\infty} \mathbb{P}(N_t = n)\mathbb{P}(S_n \in B)$$

by independence of the two events. For $k = 1$, $0 \leqslant t_0 < t_1$:

$$\begin{aligned}\mathbb{P}\left([X_{t_0} \in B_0] \cap [X_{t_1} - X_{t_0} \in B_1]\right) &= \mathbb{P}([S_{N_{t_0}} \in B_0] \cap [S_{N_{t_1}} - S_{N_{t_0}} \in B_1]) \\ &= \sum_{n_0=0}^{+\infty} \mathbb{P}([N_{t_0} = n_0] \cap [S_{n_0} \in B_0] \cap [S_{N_{t_1}} - S_{n_0} \in B_1])\end{aligned}$$

But : $S_{N_{t_1}} - S_{n_0} = \sum_{j=n_0+1}^{N_{t_1}} Z_j = \sum_{j=1}^{N_{t_1}-n_0} Z_j$ as $\text{card}(\{Z_j\}_{j\in\{n_0+1,\dots,N_{t_1}\}}) = N_{t_1} - n_0 - 1 + 1 = N_{t_1} - n_0 = \text{card}(\{Z_j\}_{j\in\{1,\dots,N_{t_1}-n_0\}})$. And using the fact $\forall p \in \mathbb{N}$, S_p is the general form of a random walk, henceforth as $(Z_n)_{n\in\mathbb{N}}$ is a sequence of i.i.d random variables : $S_{N_{t_1}-N_{t_0}} \overset{d}{=} S_{N'_{t_1-t_0}}$, where $(N't)_{t\leqslant 0}$ is another Poisson process obviously independent of $(Nt)_{t\leqslant 0}$. Therefore,

$$\begin{aligned}\mathbb{P}\left([X_{t_0} \in B_0] \cap [X_{t_1} - X_{t_0} \in B_1]\right) &= \sum_{n_0=0}^{+\infty} \mathbb{P}([N'_{t_0} = n_0] \cap [S_{n_0} \in B_0] \cap [S_{N'_{t_1}-N'_{t_0}} \in B_1]) \\ &= \sum_{n_0=0}^{+\infty} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(S_{n_0} \in B_0)\mathbb{P}(S_{N'_{t_1}-N'_{t_0}} \in B_1)\end{aligned}$$

by same logic as previously used on random walk general form :

$$= \sum_{n_0=0}^{+\infty} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(S_{n_0} \in B_0) \sum_{n_1=0}^{+\infty} \mathbb{P}(N'_{t_1} - N'_{t_0} = n_1)\mathbb{P}(S_{n_1} \in B_1)$$

$$= \sum_{n_0=0}^{+\infty} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(S_{n_0} \in B_0) \sum_{n_1=0}^{+\infty} \mathbb{P}(N'_{t_1} - N_{t_0} = n_1)\mathbb{P}(S_{n_0+n_1} - S_{n_0} \in B_1)$$

by stationary independent increment property of a Poisson process :

$$\begin{aligned}&= \sum_{n_0,n_1} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(N'_{t_1} - N'_{t_0} = n_1)\mathbb{P}(S_{n_0} \in B_0)\mathbb{P}(S_{n_0+n_1} - S_{n_0} \in B_1) \\ &= \sum_{n_0,n_1} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(N'_{t_1-t_0} = n_1)\mathbb{P}(S_{n_0} \in B_0)\mathbb{P}(S_{n_1} \in B_1) \\ &= \sum_{n_0=0}^{+\infty} \mathbb{P}(N'_{t_0} = n_0)\mathbb{P}(S_{n_0} \in B_0) \sum_{n_1=0}^{+\infty} \mathbb{P}(N'_{t_1-t_0} = n_1)\mathbb{P}(S_{n_1} \in B_1) \\ &= \mathbb{P}(X_{t_0} \in B_0)\mathbb{P}(X_{t_1-t_0} \in B_1)\end{aligned}$$

Henceforth, for $0 \leqslant t_0 < t_1 < \cdots < t_k$ we have that :

$$\begin{aligned}\mathbb{P}\left([X_{t_0} \in B_0] \cap \left(\bigcap_{i=0}^{k-1} [X_{t_{i+1}} - X_{t_i} \in B_{i+1}]\right)\right) &= \mathbb{P}(X_{t_0} \in B_0) \prod_{i=0}^{k-1} \mathbb{P}(X_{t_{i+1}-t_i} \in B_{i+1}) \\ &= \mathbb{P}(X_{t_0} \in B_0) \prod_{i=0}^{k-1} \mathbb{P}(X_{t_{i+1}} - X_{t_i} \in B_{i+1})\end{aligned}$$

So, conditions 1. and 3. are true. □

3.2 α-stable Lévy processes

We can now introduce what is a α-stable Lévy process.

Definition

Let μ be an infinitely divisible probability measure on $\mathbb{R}$. μ is called **stable** if $\forall a > 0\ \exists b > 0,\ c \in \mathbb{R}$ such that :

$$\hat{\mu}(u)^a = \hat{\mu}(bu)e^{icu} \tag{6}$$

μ is called **strictly stable** if $\forall a > 0\ \exists b > 0$ such that :

$$\hat{\mu}(u)^a = \hat{\mu}(bu) \tag{7}$$

μ is called **semi-stable** if $\forall a > 0, a \neq 1\ \exists b > 0,\ c \in \mathbb{R}$ such that (6) is verified and μ is called **strictly semi-stable** if $\forall a > 0, a \neq 1\ \exists b > 0$ such that (7) is verified.

Definition : ((Strictly) (semi-)stable Lévy process)

A Lévy process $(X_t)_{t\geqslant 0}$ is said to be stable, strictly stable, semi-stable or strictly semi-stable if $\forall t \geqslant 0$, X_t is respectively stable, strictly stable, semi-stable or strictly semi-stable.

3.3 Variance Gamma process

To conclude this section we can define a variance gamma process.

Definition : (VG process)

A variance gamma process is a Lévy process determined by a time change. Henceforth, a VG process can be expressed as $(X_t)_{t\geqslant 0}$ as the following :

$$X_t = \theta G_t + \sigma W_{G_t}$$

where $(G_t)_{t\geqslant 0}$ is a gamma process *i.e.* $\forall t \geqslant 0,\ G_t \sim \Gamma(at, b)$ for $a, b > 0$ and $(W_t)_{t\geqslant 0}$ is the standard Brownian motion.
Note that for the gamma law, the distribution μ reads :

$$\mu(dx) = x^{at-1}\frac{b^{at}}{\Gamma(at)}e^{-bx}\mathbb{1}_{\{x>0\}}(x)dx.$$

θ is called drift parameter and σ the volatility parameter.

3.4 Comparison of jump behaviors

- **Lévy measure for a compound Poisson process**

$(X_t)_{t\geqslant 0}$ given as $\forall t \geqslant 0 \quad X_t = S_{N_t}$ where $N_t \sim \mathcal{P}(\lambda t)$

$$\phi_{X_t}(u) = \mathbb{E}(e^{iuX_t}) = \mathbb{E}(e^{iuS_{N_t}}) = \mathbb{E}\left(\sum_{n=0}^{+\infty}\mathbb{1}_{\{N_t=n\}}e^{iuS_n}\right) = \sum_{n=0}^{+\infty}\mathbb{E}\left(\mathbb{1}_{\{N_t=n\}}e^{iuS_n}\right)$$
$$= \sum_{n=0}^{+\infty}\mathbb{E}(\mathbb{1}_{\{N_t=n\}})\mathbb{E}(e^{iuS_n}) = \sum_{n=0}^{+\infty}\mathbb{P}(N_t = n)\mathbb{E}(e^{iuS_n})$$

by independence of random variables $\mathbb{1}_{\{N_t=n\}}$ and e^{iuS_n}. Therefore :

$$\begin{aligned}
\mathbb{E}(e^{iuX_t}) &= \sum_{n=0}^{+\infty} \mathbb{P}(N_t=n)\mathbb{E}(e^{iuS_n}) = \sum_{n=0}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^n}{n!}\mathbb{E}(e^{iu\sum_{j=1}^n Z_j}) \\
&= \sum_{n=0}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^n}{n!}\mathbb{E}\left(\prod_{j=1}^n e^{iuZ_j}\right) = \sum_{n=0}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^n}{n!}\prod_{j=1}^n \mathbb{E}(e^{iuZ_j}) \\
&\left\{\begin{array}{r}\text{by definition of a random walk and as } (Z_j)_{j\in\mathbb{N}^*} \text{ are i.i.d. } \forall j\in\mathbb{N}^* \\ e^{iuZ_j} \overset{d}{=} e^{iuZ_1}\end{array}\right. \\
&= \sum_{n=0}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^n}{n!}(\hat{\sigma}(u))^n \quad \text{where} \quad \hat{\sigma}(u) = \mathbb{E}(e^{iuZ_1}) \\
&= e^{-\lambda t}\sum_{n=0}^{+\infty}\frac{(\lambda t\hat{\sigma}(u))^n}{n!} \\
&= e^{-\lambda t}e^{\lambda t\hat{\sigma}(u)} \\
&= e^{\lambda t(\hat{\sigma}(u)-1)}
\end{aligned}$$

As $\hat{\sigma}(u) = \mathbb{E}(e^{iuZ_1}) = \int_{\mathbb{R}} e^{iux}f(x)dx = \int_{\mathbb{R}} e^{iux}d\mathbb{P}_{Z_1}(x)$:

$$\begin{aligned}
\phi_{X_t}(u) &= \exp(\lambda t(\hat{\sigma}(u)-1)) = \exp\left(\lambda t\left(\int_{\mathbb{R}} e^{iux}f(x)dx - 1\right)\right) \\
&= \exp\left(\lambda t\int_{\mathbb{R}}(e^{iux}-1)f(x)dx\right) \\
&= \exp\left(t\int_{\mathbb{R}}(e^{iux}-1)\underbrace{\lambda f(x)dx}_{\nu(dx)}\right)
\end{aligned}$$

Hereforth, we find the form of (5) for characteristic function of compound Poisson process which is a Lévy process so X_t is infinitely divisible $\forall t \geqslant 0$. We identify the Lévy measure: $\nu = \lambda f$, where f is thus called the jump density (being the density of random variable Z_1 in definition of our chosen random walk). Here the activity of this Lévy process is finite as :

$$\int_{\mathbb{R}}\nu(dx) = \int_{\mathbb{R}}\lambda f(x)dx = \lambda\int_{\mathbb{R}} f(x)dx \underbrace{=}_{\text{by def. of a density}} \lambda$$

i.e. the number of jumps is finite with intensity of jumps λ.

Remark : λ is the expected number of jumps for this particular Lévy process.

- **Lévy measure for an α-stable process.**

An α-stable distribution is difficult to express due to analytical considerations, a particular case of such a law is developed in Sato's book :
Consider the one-sided strictly $\frac{1}{2}$-stable distribution. Its probability measure is given by :

$$\mu(dx) = \frac{c}{\sqrt{2\pi}}e^{-\frac{c^2}{2x}}x^{-\frac{3}{2}}\mathbb{1}_{\mathbb{R}_+^*}(x)dx$$

Therefore, one can express its Laplace transform. For $v > 0$:

$L_\mu(v) = \int_{\mathbb{R}_+^*} e^{-vx}\mu(dx) = \frac{c}{\sqrt{2\pi}} \int_{\mathbb{R}_+^*} e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{3}{2}} dx$ — setting $\frac{c^2}{2x} = vy$ and differentiating as a parametric integral

• For a.e. $x \in \mathbb{R}_+$, $v \mapsto e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{3}{2}}$ is $\mathcal{C}^1$ on $\mathbb{R}_+^*$.
• $\forall v > 0$, $x \mapsto e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{3}{2}}$ is piecewise continuous and integrable over $\mathbb{R}_+^*$ as per exponential criteria.
$x \mapsto -xe^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{3}{2}} = -e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{1}{2}}$ is piecewise continuous over $\mathbb{R}_+^*$.
• For a.e. $x \geqslant 0$, $\forall v > 0 \quad |-e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{1}{2}}| \leqslant e^{-\frac{c^2}{2x}} x^{-\frac{1}{2}}$ and $x \mapsto e^{-\frac{c^2}{2x}} x^{-\frac{1}{2}}$ is piecewise continuous and integrable over $\mathbb{R}_+^*$. So, by regularity of parameterized integrals theorem :

$$L'_\mu(v) = \frac{-c}{\sqrt{2\pi}} \int_{\mathbb{R}_+^*} e^{-vx} e^{-\frac{c^2}{2x}} x^{-\frac{1}{2}} dx \qquad \left(v\text{-substitution : } vx = \frac{c^2}{2y} \implies dx = -\frac{c^2}{2vy^2} dy\right)$$

using the change of variables and rearranging the integral limits

$$\left\{ = \frac{-c}{\sqrt{2\pi}} \int_{+\infty}^{0} e^{-\frac{c^2}{2y}} e^{-vy} \left(\frac{c^2}{2vy}\right)^{-\frac{1}{2}} \left(-\frac{c^2}{2vy^2}\right) dy \right.$$

$$= \frac{-c^2}{\sqrt{4\pi v}} \int_0^{+\infty} e^{-vy - \frac{c^2}{2y}} y^{-3/2} dy$$

$$= \frac{-c}{\sqrt{2v}} L_\mu(v)$$

as $L_\mu(0) = 1$ and is continuous, by definition of the Laplace transform. We are solving a 1st order linear differential equation. So:

$$L_\mu(v) = e^{-c\sqrt{2v}}$$

As a result, one can write L_μ in a different way, in order to make *Lévy-Khintchine* formulation appear, first notice the following :

$$\begin{aligned}
\int_0^{+\infty} (e^{-ux} - 1) x^{-\frac{3}{2}} dx &= -\int_0^{+\infty} x^{-\frac{3}{2}} \int_0^x ue^{-uy} dy\, dx \\
&= -\int_0^{+\infty} \int_0^x x^{-\frac{3}{2}} ue^{-uy} dy\, dx \\
&= -\int_0^{+\infty} \int_y^{+\infty} x^{-\frac{3}{2}} ue^{-uy} dx\, dy \\
&= -\int_0^{+\infty} ue^{-uy} \left(\int_y^{+\infty} x^{-\frac{3}{2}} dx\right) dy \\
&= -\int_0^{+\infty} ue^{-uy} \left(\left[-2x^{-\frac{1}{2}}\right]_y^{+\infty}\right) dy \\
&= -\int_0^{+\infty} ue^{-uy} 2y^{-\frac{1}{2}} dy \\
&= -2u \int_0^{+\infty} y^{-\frac{1}{2}} e^{-uy} dy
\end{aligned}$$

setting $z = uy$, $dy = \frac{1}{u} dz$

$$\left\{ = -2u \int_0^{+\infty} \left(\frac{z}{u}\right)^{-\frac{1}{2}} e^{-z} \frac{1}{u} dz \right.$$

y	0	$+\infty$
z	0	$+\infty$

$$= -\frac{2u}{\sqrt{u}} \int_0^{+\infty} z^{\frac{1}{2}-1} e^{-z} dz$$

$$= -2\sqrt{u}\, \Gamma\left(\frac{1}{2}\right) = -2\sqrt{\pi u}$$

so $-2\sqrt{\pi u} = \sqrt{2u} \times \sqrt{2\pi}$ *i.e.* $L_\mu(v) = \exp\left(\frac{c}{\sqrt{2\pi}} \int_0^{+\infty} (e^{-vx} - 1) x^{-\frac{3}{2}}\, dx\right)$

Finally, by extending this to the left half plane: for $v = iu$:

$$\begin{aligned} L_\mu(iu) = \hat{\mu}(u) = \mathbb{E}(e^{iux}) &= \exp\left(\frac{c}{\sqrt{2\pi}} \int_0^{+\infty} (e^{iux} - 1) x^{-\frac{3}{2}}\, dx\right) \\ &= \exp\left(\int_{\mathbb{R}} (e^{iux} - 1) \frac{c}{\sqrt{2\pi}} x^{-\frac{3}{2}} \mathbb{1}_{\{x>0\}}(x)\, dx\right) \end{aligned}$$

Thus, $\nu(dx) = \frac{c}{\sqrt{2\pi}} \mathbb{1}_{\{x>0\}}(x) x^{-\frac{3}{2}}\, dx$ and :

- $\int_{\mathbb{R}} \nu(dx) = \frac{c}{\sqrt{2\pi}} \int_0^{+\infty} x^{-\frac{3}{2}}\, dx = +\infty$ as $x \mapsto x^{-\frac{3}{2}}$ is not integrable near zero, yielding infinite activity for this law.

In general, an α-stable process admits infinitely many jumps of small magnitude, as the Lévy measure takes the form $\nu(dx) = \frac{C}{|x|^{1+\alpha}}\, dx$ for $C \in \mathbb{R}$.

- **Lévy measure for a variance gamma process.**

Let $(X_t)_{t \geqslant 0}$ be a VG process. Let $t \geqslant 0$ and $\theta \in \mathbb{R}, \sigma > 0$:

$$\phi_{X_t}(u) = \mathbb{E}(e^{iuX_t}) = \mathbb{E}(e^{iu(\theta G_t + \sigma W_{G_t})}) = \mathbb{E}\left(\mathbb{E}^{G_t}\left(e^{iu(\theta G_t + \sigma W_{G_t})}\right)\right)$$

as W_{G_t} is a Brownian motion $W_{G_t} \sim \mathcal{N}(0, G_t)$, one has, seeing G_t as constant as we are conditionally expecting by the latter, whence $\sigma W_{G_t} + \theta G_t \sim \mathcal{N}(\theta G_t, \sigma^2 G_t)$ similarly to the Gaussian technique and :

$$\mathbb{E}^{G_t}\left(e^{iu(\theta G_t + \sigma W_{G_t})}\right) = e^{iu\theta G_t - \frac{1}{2}u^2\sigma^2 G_t} = e^{(iu\theta - \frac{u^2\sigma^2}{2})G_t}$$

In addition, as $G_t \sim \Gamma(at, b)$ where $a = \frac{t}{\rho}$, $b = \rho$, for $\rho > 0$, so it yields :

$$\mathrm{E}\big(\mathbb{E}^{G_t}(e^{iu(\theta G_t + \sigma W_{G_t})})\big) = \underbrace{\mathbb{E}\left(e^{(iu\theta - \frac{u^2}{2}\sigma^2)G_t}\right) = \left(1 - \rho(iu\theta - \frac{u^2}{2}\sigma^2)\right)^{-\frac{t}{\rho}}}_{\substack{\text{by same approach} \\ \text{as for this case} \\ \text{computation in a previous} \\ \text{example.}}}$$

Using the fact that :

$$\ln(1 - iz) = -\int_0^{+\infty} (e^{izx} - 1) \frac{e^{-x}}{x}\, dx \tag{8}$$

$$\begin{aligned} 1 - iu\nu\theta + \frac{u^2\sigma^2\nu}{2} &= (1 - iu\eta_1)(1 + iu\eta_2) \\ &= 1 - iu\eta_1 + iu\eta_2 + u^2\eta_1\eta_2 \\ &= 1 - iu(\eta_1 - \eta_2) + u^2\eta_1\eta_2 \end{aligned}$$

By identification : $\begin{cases} \eta_1 - \eta_2 = \nu\theta \\ \eta_1\eta_2 = \frac{\sigma^2\nu}{2} \end{cases} \implies \begin{cases} \eta_1 = \nu\theta + \eta_2 \\ (\nu\theta + \eta_2)\eta_2 = \frac{\sigma^2\nu}{2} \end{cases}$

$$\implies \begin{cases} \eta_1 = \nu\theta + \eta_2 \\ \eta_2^2 + \nu\theta\eta_2 - \frac{\sigma^2\nu}{2} = 0 \end{cases}$$

$$\implies \eta_2 = \frac{-\nu\theta \pm \sqrt{\nu^2\theta^2 + 2\sigma^2\nu}}{2} = \frac{-\nu\theta \pm \sqrt{\nu^2\theta^2 + 2\sigma^2\nu}}{2}$$

Therefore, one either has the possibilities :

$$\begin{cases} \eta_1 = \frac{\nu\theta+\sqrt{\nu^2\theta^2+2\sigma^2\nu}}{2} \\ \eta_2 = \frac{-\nu\theta+\sqrt{\nu^2\theta^2+2\sigma^2\nu}}{2} \end{cases} \quad \text{or} \quad \begin{cases} \eta_1 = \frac{\nu\theta-\sqrt{\nu^2\theta^2+2\sigma^2\nu}}{2} \\ \eta_2 = \frac{-\nu\theta-\sqrt{\nu^2\theta^2+2\sigma^2\nu}}{2} \end{cases}$$

Either way, for simple considerations we get :

$$\begin{aligned} 1 - iu\nu\theta + \frac{u^2\sigma^2\nu}{2} &= (1 - iu\eta_1)(1 + iu\eta_2) \\ \phi_{X_t}(u) &= \exp\left(\ln\left((1 - iu\nu\theta + \frac{u^2\sigma^2\nu}{2})^{-\frac{t}{\nu}}\right)\right) \\ &= \exp\left(t\left[-\frac{1}{\nu}\ln(1 - iu\nu\theta + \frac{u^2\sigma^2\nu}{2})\right]\right) \end{aligned}$$

So according to form (5) one has the Lévy exponent $\psi(u)$ given by: $\psi(u) = -\frac{1}{\nu}\ln(1 - iu\nu\theta + \frac{u^2\sigma^2\nu}{2})$.
So by identity (8) :

$$\begin{aligned} \psi(u) &= -\frac{1}{\nu}\ln((1 - iu\eta_1)(1 + iu\eta_2)) \\ &= -\frac{1}{\nu}\left(\ln(1 - iu\eta_1) + \ln(1 + iu\eta_2)\right) \\ &= -\frac{1}{\nu}\left(-\int_0^{+\infty}(e^{iu\eta_1 x} - 1)\frac{e^{-x}}{x}\,dx - \int_0^{+\infty}(e^{-iu\eta_2 x} - 1)\frac{e^{-x}}{x}\,dx\right) \\ &= \frac{1}{\nu}\left(\int_0^{+\infty}(e^{iu\eta_1 x} - 1)\frac{e^{-x}}{x}\,dx + \int_0^{+\infty}(e^{-iu\eta_2 x} - 1)\frac{e^{-x}}{x}\,dx\right) \end{aligned}$$

$$\begin{array}{r} y = \eta_1 x \; y = -\eta_2 x \\ dx = \frac{1}{\eta_1}dy \; dx = -\frac{1}{\eta_2}dy \end{array} \left\{ \begin{aligned} &= \frac{1}{\nu}\left(\int_0^{+\infty}(e^{iuy} - 1)\frac{e^{-\frac{y}{\eta_1}}}{\frac{y}{\eta_1}}\frac{1}{\eta_1}\,dy + \int_0^{-\infty}(e^{iuy} - 1)\frac{e^{\frac{y}{\eta_2}}}{-\frac{y}{\eta_2}}\frac{1}{-\eta_2}\,dy\right) \end{aligned} \right.$$

x	0	$+\infty$
y	0	$+\infty$

x	0	$+\infty$
y	0	$-\infty$

Thus :

$$\begin{aligned} \psi(u) &= \frac{1}{\nu}\left(\int_0^{+\infty}(e^{iuy} - 1)\frac{e^{-\frac{y}{\eta_1}}}{y}\,dy + \int_{-\infty}^{0}(e^{iuy} - 1)\frac{-e^{\frac{y}{\eta_2}}}{y}\,dy\right) \\ &= \frac{1}{\nu}\int_{-\infty}^{+\infty}(e^{iuy} - 1)\frac{\left(e^{-\frac{y}{\eta_1}} - e^{\frac{y}{\eta_2}}\right)}{y}\,dy \\ &= \int_{\mathbb{R}}(e^{iuy} - 1)\underbrace{\frac{e^{-\frac{y}{\eta_1}} - e^{\frac{y}{\eta_2}}}{\nu y}\,dy}_{\nu(dy)} \end{aligned}$$

Hence, the Lévy measure for this process reads :

$$\nu(dx) = \frac{e^{-\frac{x}{\eta_1}} - e^{\frac{x}{\eta_2}}}{\nu x}\,dx$$

For this reason, $\int_{\mathbb{R}}\nu(dx) = +\infty$ as the part in $\frac{1}{x}$ blows up near 0 even with a relevant choice for constants η_1, η_2. Which is why a variance gamma has also infinitely many small jumps, *i.e.*, has infinite activity.

Remark : Parameters θ, σ and ν are respectively called drift, volatility, and variance rate of the process.

4 Approximation of Small Jumps

4.1 Simulation Issues for Infinite-Activity Lévy Processes

4.1.1 Impossibility of exact simulation

We have seen that some Lévy processes have infinite activity, meaning infinitely many small jumps occur which does leave computational methods untouched. Indeed, it becomes difficult for computers to simulate an infinite mass near 0. This case is, namely, when $\int_{|x|<1} \nu(dx) = +\infty$ where small jumps are referred to as previously mentioned. By *Lévy-Itô* decomposition, the yielded formula tells us that in such case the Lévy process is said to have infinitely many small jumps on any finite interval. Therefore, clever techniques are introduced to bypass such a consideration, such as truncation and approximation methods.

4.1.2 Truncation of the Lévy measure

To handle the infinite number of jumps near the origin in infinite-activity Lévy processes, one can truncate the Lévy measure at a threshold $\epsilon > 0$. This effectively bypasses the **singularity at zero** by partitioning the process into a finite-activity compound Poisson process (for jumps $|x| > \epsilon$) and a residual small-jump component. For a Lévy process $(L_t)_{t\geqslant 0}$, the truncation is formally defined using the indicator function : $\nu_\epsilon(dx) = \mathbb{1}_{\{|x|>\epsilon\}}\nu(dx)$ for $\epsilon > 0$, allowing for a tractable simulation of the significant jumps resulting in removing the mass near 0. Therefore, decomposition gives using :

$$L_t^{\epsilon,0} := L_t - \sum_{s\leqslant t} \Delta L_s \mathbb{1}_{\{|\Delta L_s|<\epsilon\}}$$

which is given for finite variation case *i.e.* $\int_{|x|<1} |x|\nu(dx) < \infty$.
For the compensated case, namely $\int_{|x|<1} |x|\nu(dx) = \infty$, one gets the following :

$$L_t^{\epsilon,1} := \mu_\epsilon t + \sqrt{c}W_t + N^\epsilon(t) \quad \text{where} \quad \mu_\epsilon = b - \int_{\epsilon\leqslant|x|<1} x\nu(dx)$$

(adding that $N^\epsilon(t) = \sum_{s\leqslant t} \Delta L_s \mathbb{1}_{\{|\Delta L_s|>\epsilon\}}$ is a Poisson process with jump measure ν_ϵ).
One can also see how to link the finite variation case to the compensated case by :

$$L_t^{\epsilon,1} = L_t^{\epsilon,0} + \mathbb{E}(L_t - L_t^{\epsilon,0})$$

4.1.3 Separation of small and large jumps

To now improve approximation, introduce :

$$L_t^{\epsilon,2} = \mu_\epsilon t + \sqrt{c + \sigma^2(\epsilon)}W_t + N^\epsilon(t) \quad \text{where} \quad \sigma^2(\epsilon) := \int_{|x|<\epsilon} x^2\nu(dx)$$

Here, it is noticeable that a Gaussian term appears in the second term which yields the error defined by : $L_t^\epsilon := L_t - L_t^{\epsilon,2}$ to have a Gaussian approximation which goes in the direction of the truncated expression of a Lévy process $(L_t)_{t\geqslant 0}$ according to the *Lévy-Itô* decomposition adjusted up to a cutoff $\epsilon > 0$:

$$L_t = bt + \sqrt{c}W_t + \underbrace{\int_{|x|\geqslant\epsilon} xN(t,dx)}_{X_t^\epsilon} + \underbrace{\int_{|x|<\epsilon} x\tilde{N}(t,dx)}_{R_t^\epsilon}$$

One can see that: $N(t,dx) = \int_0^t \mu^L(ds,dx)$ is the usual Poisson random measure defined for the decomposition. And this analysis is equivalent to saying that small jumps are replaced by a Gaussian term being :

$$R_t^\epsilon = \int_{|x|<\epsilon} x\tilde{N}(t,dx) \quad \longleftrightarrow \quad \sigma(\epsilon)W_t' \qquad (W_t')_{t\geqslant 0} \quad \text{Brownian motion}$$

Remark : One can pick up that : $N^\epsilon(t) = \int_{|x|\geqslant\epsilon} xN(t,dx)$ which gives exactly the expression $L_t^{\epsilon,1}$.

$$\begin{aligned} L_t - L_t^{\epsilon,1} &= bt + \cancel{\sqrt{\epsilon}W_t} + \cancel{\int_{|x|\geqslant\epsilon} xN(t,dx)} + \int_{|x|<\epsilon} x\tilde{N}(t,dx) - (\mu_\epsilon t + \cancel{\sqrt{\epsilon}W_t} + \cancel{N^\epsilon(t)}) \\ &= \cancel{bt} - \cancel{bt} + \int_{\epsilon\leqslant|x|<1} |x|\nu(dx) + \int_{|x|<\epsilon} x\tilde{N}(t,dx) \\ &= \int_{|x|\geqslant\epsilon} x\nu(dx) + \int_{\epsilon\leqslant|x|<1} |x|\nu(dx) \end{aligned}$$

4.2 Gaussian Approximation of Small Jumps

4.2.1 Heuristic motivation

The heuristic motivation for this approximation lies in the decomposition of the Lévy process into two independent components based on a jump-size threshold $\epsilon > 0$. While the large-jump component X_t^ϵ consists of a finite number of jumps and can be simulated exactly as a compound Poisson process, the small-jump residual R_t^ϵ poses a numerical challenge due to its infinite activity. However, since R_t^ϵ represents the sum of an infinite number of independent, infinitesimal increments, the Central Limit Theorem suggests that this component will asymptotically behave like a Gaussian process. Consequently, R_t^ϵ can be effectively modeled as a "cloud of Gaussian noise," significantly simplifying the simulation while preserving the statistical properties of the original process.

4.2.2 Central Limit Effects

The transition from a jump-based process to a continuous approximation is justified by the **Central Limit Theorem (CLT)** applied to the small-jump component R_t^ϵ. Since an infinite-activity Lévy process contains an infinite number of jumps in any finite time interval $[0,t]$, the residual R_t^ϵ can be viewed as the sum of a large number of independent, infinitesimal random variables. In fact, R_t^ϵ represents a stochastic integral of the small jumps.

$$R_{\Delta t}^\epsilon \approx \sum_{i=1}^{N} Y_i$$

where:

- N is the (infinite or very large) number of jumps occurring in the interval ($[0,t]$).
- Y_i are the individual jump sizes, which are **independent and identically distributed (i.i.d.) random variables** (once you condition on the number of jumps, similar to what we saw in the case of compound Poisson process where jump size is given by the random variable scheme in the random walk).

As the threshold ϵ decreases, the frequency of these jumps increases, while their individual magnitudes decrease. If the variance of these jumps, denoted by $\sigma^2(\epsilon)$, is finite, then the properly centered and scaled sum of these increments converges in distribution to a normal distribution. Formally, as $\epsilon \to 0$, the fluctuations of R_t^ϵ around its mean become indistinguishable from the paths of a Brownian motion. This "averaging out" effect allows us to replace the computationally expensive simulation of infinite tiny jumps with a single Gaussian random variable, provided the Lévy measure is sufficiently "dense" near the origin to trigger these central limit effects.

4.2.3 Construction of the Gaussian approximation

The practical construction of the approximation involves replacing the small-jump component R_t^ϵ with a continuous Gaussian process that matches its infinitesimal mean and variance. Given a truncation threshold $\epsilon > 0$, we define the volatility of the approximating Brownian motion as :

$$\sigma^2(\epsilon) = \int_{|x|<\epsilon} x^2 \nu(dx)$$

and the corresponding drift correction as :

$$\mu(\epsilon) = \int_{|x|<\epsilon} x\nu(dx) = \mu_\epsilon - \gamma$$

The approximated Lévy process $X_t \approx \hat{X}_t$ is then constructed as the sum of three independent parts :

1. A **drift component** capturing the mean of the small jumps and the original model drift.
2. A **diffusion component** $\sigma(\epsilon)W_t$, where W_t is a standard Brownian motion.
3. A **jump component** simulated as a Compound Poisson process for all jumps $|x| > \epsilon$.

This construction ensures that the first two moments of the approximated process remain consistent with the original infinite-activity model.

4.3 Asmussen & Rosiński (2001)

4.3.1 Main approximation results

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{P})$ be a filtered space.
By the previously defined error, Asmussen and Rosiński have thought a way to approximate what corresponds to the small jumps component. So much that, $(L^\varepsilon)_{t\geqslant 0}$ is seen as Lévy process as well with characteristic function given by :

$$\phi_{L_t^\varepsilon}(u) = \mathbb{E}(e^{iuL_t^\varepsilon}) = \exp\left(t\int_{|x|<\varepsilon} (e^{iux} - 1 - iux)\,\nu(dx)\right) \quad \text{of the form (5)}$$

Firstly, define the following space of functions :

Definition : (Skorokhod space : càdlàg functions)

Let (M, d) be a metric space and $E \subseteq \mathbb{R}$. A function $f : E \longrightarrow M$ is said to be **càdlàg** if $\forall\ t \in E$:

- $f(t-) := \lim_{\substack{s\to t \\ s<t}} f(s)$ left limit exists
- $f(t+) := \lim_{\substack{s\to t \\ s>t}} f(s)$ right limit exists and is equal to $f(t)$.

(Usual limit definition with respect to the metric *i.e.* $f(\alpha) \xrightarrow{\alpha\to\gamma} \beta$ if and only if $d(f(\alpha), \beta) \xrightarrow{\alpha\to\gamma} 0$).

Then,

$$\frac{L^\varepsilon}{\sigma(\varepsilon)} \xrightarrow[\varepsilon\to 0]{d} W \iff \forall\ \varphi : D([0,1]) \longrightarrow \mathbb{R} \text{ continuous, bounded, measurable} \quad \mathbb{E}\left(\varphi\left(\frac{L^\varepsilon}{\sigma(\varepsilon)}\right)\right) \xrightarrow{\varepsilon\to 0} \mathbb{E}(\varphi(W))$$

Combining these ideas, the main approximation theorem rises from this :

Theorem :

$$\frac{L^\varepsilon}{\sigma(\varepsilon)} \xrightarrow[\varepsilon\to 0]{d} W \iff \forall\ \kappa > 0 \quad \sigma(\min(\kappa\sigma(\varepsilon), \varepsilon)) \underset{\varepsilon\to 0}{\sim} \sigma(\varepsilon) \tag{9}$$

To get a grip on the proof of this result, a few notions need to be introduced, such as :
For a filtration $\mathcal{F} = (\mathcal{F}_t)_{t\geqslant 0}$, a random variable τ taking values in $\overline{\mathbb{R}_+}$ is said to be :

- a **stopping time** if $\forall\ t \geqslant 0,\ \{\tau \leqslant t\} \in \mathcal{F}_t$ as an event. (In other words, at any time t one knows for sure if the event has already happened.)
- an **optional time** if $\forall\ t > 0,\ \{\tau < t\} \in \mathcal{F}_t$ as an event. (In other words, at time t, one only knows if the event has happened strictly before now.)

Results from Kallenberg's book will be used in the proof :

Theorem : (Strong Markov property)

For any $\mathcal{F}$-Brownian motion B in Ω and a.s. finite $\mathcal{F}^+$-optional time τ, the process $B'_t = B_{\tau+t} - B_\tau\ \forall\ t \geqslant 0$ is again a Brownian motion $\perp \mathcal{F}_\tau^+$ where :

$$\mathcal{F}_\tau = \{A \in \mathcal{F} \mid A \cap \{\tau \leqslant t\} \in \mathcal{F}_t,\ t \geqslant 0\} \quad \text{and} \quad \mathcal{F}_{\tau^+} : \mathcal{F}_\tau^+ = \{A \in \mathcal{F} \mid A \cap \{\tau < t\} \in \mathcal{F}_t,\ t > 0\} = \bigcap_{u>\tau} \mathcal{F}_u$$

Remark : This property means that, for a Brownian motion, everything happening before the optional time (or Markov time) is not relevant, as stopping at time τ is equivalent to starting a new process at point B_τ.

Lemma : (Reflection principle)

Let B be a Brownian motion and τ an associated optional time τ. Then: $B \overset{d}{=} \widetilde{B}$ where
$\widetilde{B}_t = B_{\min(t,\tau)} - (B_t - B_{\min(t,\tau)})$

Proof : It suffices to compare distributions up to fixed time t, assume $\tau < +\infty$. Defining $B_t^\tau = B_{\min(t,\tau)}$ and $B'_r = B_{\tau+r} - B_\tau$, the previous theorem tells us that $B' = (B'_r)_{r\geqslant 0}$ is a Brownian motion and B' is independent of (τ, B^τ). Moreover, since $-B' \overset{d}{=} B' \implies (\tau, B^\tau, B') \overset{d}{=} (\tau, B^\tau, -B')$.
And $B_r = B_r^\tau + B'_{\max(0,r-\tau)}$ with $\widetilde{B}_r = B_r^\tau - B'_{\max(0,r-\tau)}$ for $r \geqslant 0$. Henceforth :

$$\widetilde{B}_t = B_t^\tau - B'_{\max(0,t-\tau)} = B_t^\tau - (B_t^\tau + B'_{\max(0,t-\tau)} - B_t^\tau) = B_t^\tau - (B_t - B_t^\tau)$$

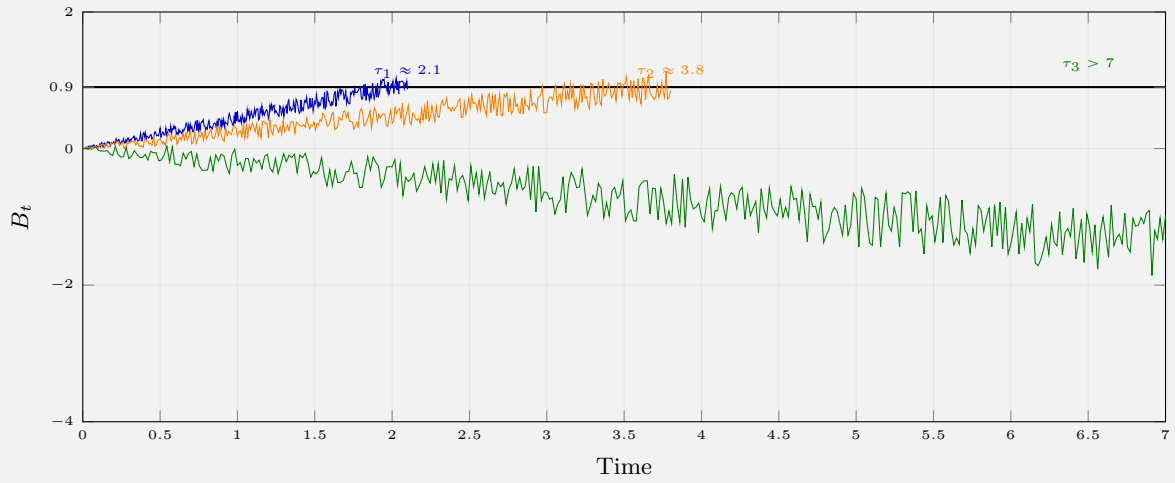
□

Remark : Types of times associated to a process and that are said to be optional are listed below :

Type of time	**Status**	**Definition**
Debuts of a progressive set	optional	$\tau = \inf\{t : (t, X_t) \in A\}$
First entry time (closed)	stopping (thus optional)	$\tau = \inf_{t\geqslant 0}\{X_t \in [a, b]\}$
First entry time (open)	optional	$\tau = \inf_{t\geqslant 0}\{X_t \in]a, b[\}$

Example :

Here is an example of stopping time which is a hitting time of a Brownian motion. One can see that the process begins at 0 and stops at 1 showing various values taken by τ.



4.3.2 Assumptions on the Lévy measure

Additional results stated below will be used to prove main approximation theorem.

A family of random vectors $(\xi_t)_{t\in T} \subset \mathbb{R}^d$ is said to be **tight** if

$$\lim_{r\to\infty} \sup_{t\in T} \mathbb{P}(|\xi_t| > r) = 0$$

For a sequence $(\xi_n)_n$ this is equivalent to :

$$\lim_{r\to\infty} \limsup_{n\to\infty} \mathbb{P}(|\xi_n| > r) = 0$$

Lemma 4.8 : (Weak convergence and tightness)

Let $\xi, (\xi_n)_n$ be random vectors in $\mathbb{R}^d$ verifying $\xi_n \xrightarrow{d} \xi$. Then (ξ_n) is tight.

Proof : $\forall\ r > 0$, define $f(x) = \min(0, 1 - \min(0, r - |x|))$. Then :

$$\limsup_{n\to\infty} \mathbb{P}(|\xi_n| > r) \leqslant \lim_{n\to\infty} \mathbb{E}(f(\xi_n)) = \mathbb{E}(f(\xi)) \leqslant \mathbb{P}(|\xi| > r - 1) \xrightarrow{r\to\infty} 0$$

□

Lemma 5.1 : (Tail estimates)

For any probability measure μ on $\mathbb{R}$, we have : for $r > 0$

$$\mu(\{|x| > r\}) \leqslant \frac{r}{2} \int_{-2/r}^{2/r} (1 - \hat{\mu}(t))\, dt \tag{10}$$

$$\mu([-r, r]) \leqslant 2r \int_{-2/r}^{2/r} |\hat{\mu}(t)|\, dt \tag{11}$$

Proof : For $x \geqslant 2,\ \sin(x) \leqslant \frac{x}{2}$, by Fubini's theorem, $\forall\ c > 0$:

$$\begin{aligned}
\int_{-c}^{c} (1 - \hat{\mu}(t))\, dt &= \int_{-c}^{c} \left(1 - \int e^{itx} \mu(dx)\right) dt \\
&= \int_{-c}^{c} \left(\int \mu(dx) - \int e^{itx} \mu(dx)\right) dt \\
&= \int_{-c}^{c} \int (1 - e^{itx})\, \mu(dx)\, dt \\
&= \int \mu(dx) \int_{-c}^{c} (1 - e^{itx})\, dt \\
&= \int \mu(dx) \left(2c - \left[\frac{e^{itx}}{ix}\right]_{-c}^{c}\right) \\
&= \int \mu(dx) \left(2c - \left(\frac{e^{icx}}{ix} - \frac{e^{-icx}}{ix}\right)\right)
\end{aligned}$$

$$\begin{aligned}
&= \int \mu(dx) \left(2c - \frac{2\sin(cx)}{x}\right) = 2c \int \left(1 - \frac{\sin(cx)}{cx}\right) \mu(dx) \\
&\geqslant 2c \int_{\{|cx| \geqslant 2\}} \left(1 - \frac{\sin(cx)}{cx}\right) \mu(dx) \\
&\geqslant 2c \int_{\{|cx| \geqslant 2\}} \left(1 - \frac{1}{2}\right) \mu(dx) = c \int \mathbb{1}_{\{|cx| \geqslant 2\}}(x) \mu(dx) = c\mu(\{|cx| \geqslant 2\})
\end{aligned}$$

By setting $c = \frac{2}{r}$, one gets :

$$\frac{2}{r}\mu\left(\left\{\left|\frac{2}{r}x\right| \geqslant 2\right\}\right) \leqslant \int_{-2/r}^{2/r} (1 - \hat{\mu}(t))\, dt \quad i.e. \quad \mu(\{|x| \geqslant r\}) \leqslant \frac{r}{2} \int_{-2/r}^{2/r} (1 - \hat{\mu}(t))\, dt$$

which proves (10).
Furthermore, notice that : for any w, s.t. $|w| \leqslant 1,\ \frac{1}{2} \leqslant 2\frac{1-\cos(w)}{w^2}$. So for $x \in [-r, r]$, this is true :

$$\frac{1}{2}\mu([-r, r]) = \frac{1}{2} \int_{-r}^{r} \mu(dx) \leqslant 2 \int_{-r}^{r} \frac{1 - \cos(x/r)}{(x/r)^2} \mu(dx) \quad \text{as } \frac{1}{2} \leqslant 2\frac{1 - \cos(x/r)}{(x/r)^2}$$

Hence : $\frac{1}{2}\mu([-r, r]) \leqslant 2\int_{\mathbb{R}} \frac{1-\cos(x/r)}{(x/r)^2} \mu(dx)$ and using the fact that :

$$2\frac{1 - \cos(w)}{w^2} = \int_{-1}^{1} (1 - |t|) e^{itw}\, dt \quad \text{so} \quad 2\frac{1 - \cos(x/r)}{(x/r)^2} = \int_{-1}^{1} (1 - |t|) e^{itx/r}\, dt = r \int_{-1/r}^{1/r} (1 - r|t|) e^{itx}\, dt$$

Now :

$$\begin{aligned}
\frac{1}{2}\mu([-r, r]) &\leqslant \int_{\mathbb{R}} \left(\int_{-1/r}^{1/r} (1 - r|t|) e^{itx} r\, dt\right) \mu(dx) \\
&= r \int_{-1/r}^{1/r} (1 - r|t|) \left(\int e^{itx} \mu(dx)\right) dt = r \int_{-1/r}^{1/r} (1 - r|t|) \hat{\mu}(t)\, dt \\
&\leqslant r \int_{-1/r}^{1/r} |\hat{\mu}(t)|\, dt
\end{aligned}$$

i.e. : $\mu([-r, r]) \leqslant 2r \int_{-1/r}^{1/r} |\hat{\mu}(t)|\, dt$ being (11).

□

Remark : A family of characteristic functions $(\hat{\mu}_\alpha)_\alpha$ is said to be **equicontinuous at 0** if $\forall\ \varepsilon > 0, \exists\ \delta > 0$ s.t.

$|t| < \delta \implies |1 - \hat{\mu}_\alpha(t)| < \varepsilon \ \forall\ \alpha$.
And that $(\mu_\alpha)_\alpha$ is **tight** if $\lim_{r\to\infty} \sup_\alpha \mu_\alpha(\{|x| > r\}) = 0$.

Lemma 5.2 : (Equicontinuity and tightness)

A family $(\mu_\alpha)_\alpha$ of distributions is **tight iff** $(\hat{\mu}_\alpha)_\alpha$ is **equicontinuous at 0**, thus $(\hat{\mu}_\alpha)_\alpha$ is **uniformly equicontinuous**.

Proof :
$\boxed{\Longleftarrow}$ Suppose $(\hat{\mu}_\alpha)_\alpha$ to be equicontinuous, then by the previous lemma tightness of (μ_α) is immediate.
Let ξ_α be a random vector with probability measure μ_α. $\forall\ s, t \in \mathbb{R}$:

$$\begin{aligned}|\hat{\mu}_\alpha(s) - \hat{\mu}_\alpha(t)| &\leqslant \mathbb{E}(|e^{is\xi_\alpha} - e^{it\xi_\alpha}|) = \mathbb{E}(|e^{is\xi_\alpha}(1 - e^{i(t-s)\xi_\alpha})|) \\ &= \underbrace{|\mathbb{E}(|1 \cdot e^{is\xi_\alpha}|)|}_{=1}\, \mathbb{E}(|1 - e^{i(t-s)\xi_\alpha}|) = \mathbb{E}(|1 - e^{i(t-s)\xi_\alpha}|) \\ &\leqslant 2\mathbb{E}(\min(|(t-s)\xi_\alpha|, 1)) \quad (*)\end{aligned}$$

$\boxed{\Longrightarrow}$ Suppose $(\mu_\alpha)_\alpha$ to be tight, then $\mathbb{E}(\min(|(t-s)\xi_\alpha|, 1)) \xrightarrow{t-s\to 0} 0$ uniformly in α. Thus uniform equicontinuity yields as taking $t - s \to 0$ in $(*)$ is equivalent to writing :
For $\varepsilon > 0$, let $\delta > 0$ s.t. $|t - s| < \delta$ then $|\hat{\mu}_\alpha(t) - \hat{\mu}_\alpha(s)| \leqslant 2\mathbb{E}(\min(|(t-s)\xi_\alpha|, 1)) < \varepsilon$.

□

Lemma 5.3 : (Uniqueness and continuity)

For any probability measure μ and sequence $(\mu_n)_{n\in\mathbb{N}}$, $\mu_n \xrightarrow{w} \mu \iff \hat{\mu}_n(t) \to \hat{\mu}(t)\ \forall\ t$.
Then $\hat{\mu}_n \to \hat{\mu}$ uniformly on every bounded set.

Lemma 5.4 : (Consequence of Stone-Weierstrass approximation)

Any $f : \mathbb{R} \to \mathbb{R}$ continuous 2π-periodic admits a uniform approximation by linear combinations of $\cos(kx)$ and $\sin(kx)$ for $k \in \mathbb{N}$.

Proof : (Lemma 5.3)
$\boxed{\Longrightarrow}$ If $\mu_n \xrightarrow{w} \mu$ then $\hat{\mu}_n(t) \to \hat{\mu}(t)$ as $x \mapsto e^{itx}$ is continuous and bounded on $\mathbb{R}\ \forall\ t$.
Moreover, this convergence is uniform on every bounded set by Lemmas 4.8 and 5.2.

$\boxed{\Longleftarrow}$ Suppose that $\hat{\mu}_n(t) \to \hat{\mu}(t)\ \forall\ t$. By Lemma 5.1 and dominated convergence $\forall\ a \in \mathbb{R}$ and some $r > 0$:

$$\limsup_{n\to\infty} \mu_n(\{|ax| > r\}) \leqslant \lim_{n\to\infty} \frac{r}{2} \int_{-2/r}^{2/r} (1 - \hat{\mu}_n(at))\, dt = \frac{r}{2} \int_{-2/r}^{2/r} (1 - \hat{\mu}(at))\, dt$$

$\hat{\mu}$ is continuous at 0 thus $\lim_{r\to\infty} \int_{-2/r}^{2/r} (1 - \hat{\mu}(at))\, dt = 0$ proving that $(\mu_n)_n$ is tight.
For $\varepsilon > 0, \exists\ r > 0$ s.t. $\mu_n(\{|x| > r\}) \leqslant \varepsilon\ \forall\ n \in \mathbb{N}$ and $\mu(\{|x| > r\}) \leqslant \varepsilon$. So for such r :
Let $f \in C_b(\mathbb{R})$, so $\exists\ m > 0$ s.t. $|f| \leqslant m$. Let $f_r = f\mathbb{1}_{\{|x|\leqslant r\}}$, extending it to $\tilde{f}$ continuous on $\mathbb{R}$ with $|\tilde{f}| \leqslant m$ and

$2\pi r$ periodic. By Lemma 5.4, $\exists\ g$ linear combinations of $\cos(\frac{kx}{r})$ and $\sin(\frac{kx}{r})$ for $k \in \mathbb{N}$ s.t. $|\tilde{f} - g| \leqslant \varepsilon\ \forall\ x \in \mathbb{R}$:

$$\begin{aligned}
|\mu_n f - \mu_n g| &= |\mu_n f - \mu_n \tilde{f} + \mu_n \tilde{f} - \mu_n g| \leqslant |\mu_n f - \mu_n \tilde{f}| + |\mu_n \tilde{f} - \mu_n g| \\
&= \left| \int f(x)\mu_n(dx) - \int \tilde{f}(x)\mu_n(dx) \right| + |\mu_n(\tilde{f} - g)| \\
&= \left| \int_{|x|\leqslant r} f(x)\mu_n(dx) + \int_{|x|>r} f(x)\mu_n(dx) - \int_{|x|\leqslant r} \tilde{f}(x)\mu_n(dx) - \int_{|x|>r} \tilde{f}(x)\mu_n(dx) \right| + |\mu_n(\tilde{f} - g)| \\
&= \left| \int_{|x|>r} (f(x) - \tilde{f}(x))\mu_n(dx) \right| + \left| \int (\tilde{f} - g)(x)\mu_n(dx) \right| \\
&\leqslant \int_{|x|>r} \|f - \tilde{f}\|_\infty \mu_n(dx) + \int \|\tilde{f} - g\|_\infty \mu_n(dx) \\
&= \mu_n(|x| > r)\|f - \tilde{f}\|_\infty + \|\tilde{f} - g\|_\infty \\
&\leqslant \mu_n(|x| > r)(\|f\|_\infty + \|\tilde{f}\|_\infty) + \|\tilde{f} - g\|_\infty \leqslant \varepsilon(m + m) + \varepsilon = (2m + 1)\varepsilon
\end{aligned}$$

and similarly for μ. Then :

$$\begin{aligned}
|\mu_n f - \mu f| &= |\mu_n f - \mu_n g + \mu_n g - \mu g + \mu g - \mu f| \\
&\leqslant |\mu_n f - \mu_n g| + |\mu_n g - \mu g| + |\mu g - \mu f| \\
&\leqslant (2m + 1)\varepsilon + |\mu_n g - \mu g| + (2m + 1)\varepsilon \\
&= |\mu_n g - \mu g| + 2(2m + 1)\varepsilon
\end{aligned}$$

Rewritten : $|\mu_n f - \mu f| \leqslant |\mu_n g - \mu g| + 2(2m + 1)\varepsilon$. Taking $n \to \infty$ we get : $|\mu_n f - \mu f| \xrightarrow{n\to\infty} 2(2m + 1)\varepsilon$.
This being true $\forall\ \varepsilon > 0$, $|\mu_n f - \mu f| \xrightarrow{n\to\infty} 0$, *i.e.*, $\mu_n f \xrightarrow{n\to\infty} \mu f$ as g is a linear combination of real parts and imaginary parts of the characteristic function evaluated at point $t = \frac{k}{r}$, so naturally $|\mu_n g - \mu g| \xrightarrow{n\to\infty} 0$ by hypothesis.
We thus have $\mu_n f \to \mu f\ \forall\ f \in C_b(\mathbb{R})$, therefore $\mu_n \xrightarrow{w} \mu$.

□

All the previous are used to prove the one below invoked in the ultimate proof of the approximation theorem.

Theorem : (Convergence of infinitely divisible distributions)

(i) Let μ be an infinitely divisible distribution with generating triplet (b, c, ν) and $(\mu_n)_n$ a sequence of infinitely divisible distributions with triplets (b_n, c_n, ν_n) on $\mathbb{R}^d\ \forall\ n \in \mathbb{N}$. Suppose : $\forall\ h > 0,\ \nu(\{|x| = h\}) = 0$, then

$$(\mu_n \xrightarrow{w} \mu) \iff (b_n^h \xrightarrow{n\to\infty} b^h,\ c_n^h \xrightarrow{n\to\infty} c^h,\ \nu_n \xrightarrow{v} \nu) \text{ on } \overline{(\mathbb{R}^d)*}$$

(ii) Let μ be an infinitely divisible distribution with generating pair (b, ν) and $(\mu_n)_n$ a sequence of infinitely divisible distributions with pair (b_n, ν_n) on $\mathbb{R}_+\ \forall\ n \in \mathbb{N}$. Suppose : $\forall\ h > 0,\ \nu(\{x = h\}) = 0$ then

$$(\mu_n \xrightarrow{w} \mu) \iff (b_n^h \xrightarrow{n\to\infty} b^h,\ \nu_n \xrightarrow{v} \nu) \text{ on } \overline{\mathbb{R}_+^*}$$

where $\mu_n \xrightarrow{w} \mu \iff \mu_n f \to \mu f\ \forall\ f \in \mathcal{C}_b(\mathbb{R})$ and $\nu_n \xrightarrow{v} \nu \iff \nu_n f \to \nu f\ \forall\ f \in \mathcal{C}_K^+$ with $\nu f = \int f d\nu$.
With $b^h = b + \int_{x\leqslant h} x\,\nu(dx)$ and $c^h = c - \int_{h\leqslant|x|\leqslant 1} x\,\nu(dx)$.

Remark : Notice that $\mathcal{C}_b(\mathbb{R}) = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ bounded continuous}\}$
and $\mathcal{C}_K^+ = \{f : \mathbb{R} \to \mathbb{R}_+ \mid f \text{ continuous with compact support}\}$.

Remark : It is possible to show that $\psi(u)$ in (5) can be rewritten :

$$\psi(u) = i\gamma u + \int_{\mathbb{R}} \left(e^{iux} - 1 - \frac{iux}{1+x^2} \right) \frac{1+x^2}{x^2} \tilde{\nu}(dx) \quad \text{with } \tilde{\nu}(dx) = c\,\delta_0(dx) + \frac{x^2}{1+x^2}\nu(dx) \tag{12}$$
$$\text{and } \gamma = b + \int \left(\frac{x}{1+x^2} - x\mathbb{1}_{\{|x|<1\}}(x) \right) \nu(dx)$$

which is useful to use in the proof.
Indeed, using the previously defined parameters :

$$\begin{aligned}
\psi(u) &= i(b + \int (\frac{x}{1+x^2} - x\mathbb{1}_{\{|x|<1\}}(x))\nu(dx))u + \int_{\mathbb{R}} (e^{iux} - 1 - \frac{iux}{1+x^2}) \frac{1+x^2}{x^2} (c\,\delta_0(dx) + \frac{x^2}{1+x^2}\nu(dx)) \\
&= ibu + iu \int (\frac{x}{1+x^2} - x\mathbb{1}_{\{|x|<1\}}(x))\nu(dx) + \int (e^{iux} - 1)\nu(dx) - \int \frac{iux}{1+x^2}\nu(dx) + \int_{\mathbb{R}} (e^{iux} - 1 - \frac{iux}{1+x^2}) \frac{1+x^2}{x^2} c\,\delta_0(dx) \\
&= ibu + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}(x))\nu(dx) + \int_{\mathbb{R}} (e^{iux} - 1 - \frac{iux}{1+x^2}) \frac{1+x^2}{x^2} c\,\delta_0(dx)
\end{aligned}$$

And : $\int_{\mathbb{R}} (e^{iux} - 1 - \frac{iux}{1+x^2}) \frac{1+x^2}{x^2} c\,\delta_0(dx) = \int_{\mathbb{R}} w(x)\delta_0(dx) = w(0)$, but as w is not defined at 0, one uses a Taylor expansion :

$$e^{iux} = 1 + iux - \frac{u^2x^2}{2} + O(x^3)$$

Plugging this in w :

$$\begin{aligned}
w(x) &= \left(1 + iux - \frac{u^2x^2}{2} + O(x^3) - 1 - \frac{iux}{1+x^2} \right) \frac{1+x^2}{x^2} c \\
&\approx_{x\to 0} \left(iux - \frac{u^2x^2}{2} - iux \right) \frac{1}{x^2} c = -\frac{u^2x^2}{2} \sigma^2 \frac{1}{x^2} \xrightarrow{x\to 0} -\frac{u^2}{2} c
\end{aligned}$$

So by continuity extension of w to $\tilde{w}$, one has $\tilde{w}(0) = \lim_{x\to 0} w(x) = -\frac{u^2}{2} c$. Hence :

$$\psi(u) = iub + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}(x))\nu(dx) + \tilde{w}(0) = iub - \frac{1}{2}u^2 c + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}_{\{|x|<1\}}(x))\nu(dx)$$

where $\tilde{w}(x) = \begin{cases} w(x) & \text{if } x \neq 0 \\ -\frac{u^2}{2} c & \text{if } x = 0 \end{cases}$

Lemma 15.15 : (One-dimensional criteria)

(i) Let $(c, \tilde{\gamma})$ associated to μ and $((c_n, \tilde{\gamma}_n))_n$ associated to $(\mu_n)_n$ on $\mathbb{R}$. Then $\mu_n \xrightarrow{w} \mu \iff \begin{cases} c_n \to c \\ \tilde{\gamma}_n \xrightarrow{w} \gamma \end{cases}$

(ii) Let $\nu, (\nu_n)_n$ Lévy measures associated with μ and μ_n on $\mathbb{R}^*$. Then $\mu_n \xrightarrow{w} \mu \iff \nu_n \xrightarrow{w} \nu$.

Proof :
(i) Defining (ψ_n) and ψ as in (12). Thus $\hat{\mu} = e^{\psi}$ and $\mu_n = e^{\psi_n}$.
$\boxed{\Longrightarrow}$ If $c_n \longrightarrow c$ and $\tilde{\nu}_n \xrightarrow{w} \tilde{\nu}$, then $\psi_n \longrightarrow \psi$ by boundedness and continuity of the integrand in (12), thus $\hat{\mu}_n \longrightarrow \hat{\mu} \implies \mu_n \xrightarrow{w} \mu$ by Lemma 5.3.
$\boxed{\Longleftarrow}$ Conversely, $\mu_n \xrightarrow{w} \mu \implies \psi_n \longrightarrow \psi$ uniformly on bounded intervals so $\psi_n \longrightarrow \psi$ in the same sense.

Introducing, very similarly as in (3), and equivalently $(\chi_n)_n$:

$$\chi(u) = \int_{-1}^{1} (\psi(u) - \psi(u+s))\, ds = 2 \int e^{iux}(1 - \sin(x)) \frac{1+x^2}{x^2} \tilde{\nu}(dx)$$

$\chi_n \longrightarrow \chi$, thus, again, by Theorem 5.3 :

$$(1 - \frac{\sin(x)}{x}) \frac{1+x^2}{x^2} \tilde{\nu}_n(dx) \xrightarrow{w} (1 - \frac{\sin(x)}{x}) \frac{1+x^2}{x^2} \tilde{\nu}(dx)$$

As the integrand in $\chi(u)$ is continuous and bounded away from 0, $\tilde{\nu}_n \xrightarrow{w} \tilde{\nu} \implies \int (e^{iux} - 1 - \frac{iux}{1+x^2}) \frac{1+x^2}{x^2} \tilde{\nu}(dx) < +\infty$ so $c_n \longrightarrow c$.
(ii) same means and using the fact that : $\lambda(u) = ua + \int_{\mathbb{R}} (1 - e^{-ux}) \tilde{\nu}(dx)$ verifies the functional relation: $\lambda(u+v) - \lambda(u) = \int e^{-ux} \tilde{\nu}(dx)$.

□

Proof : (Convergence of infinitely divisible distributions)
(ii) Suppose $(m_n)_n, m$ to be finite measures. So $(m_n \xrightarrow{w} m) \iff (m_n \xrightarrow{v} m$ on $\bar{\mathbb{R}}^*$ and $\exists h > 0$ such that $m_n([-h,h]) \longrightarrow m([-h,h])$ with $m(\{\pm h\}) = 0)$. Hence for distributions $(\mu_n)_n$ and μ on $\mathbb{R}$, $(\tilde{\nu}_n \xrightarrow{w} \tilde{\nu}) \iff (\nu_n \xrightarrow{v} \nu$ on $\bar{\mathbb{R}}^*$ and $a_n^h \longrightarrow a^h \quad \forall h > 0$ with $\nu(\{\pm h\}) = 0)$.
By the same means, $\tilde{\nu}_n \xrightarrow{w} \tilde{\nu}$ is verified for distributions $(\mu_n)_n$ and μ on $\mathbb{R}$ if and only if $\nu_n \xrightarrow{v} \nu$ on $\bar{\mathbb{R}}^*$ and $a_n^h \longrightarrow a^h \quad \forall h > 0$ with $\nu(\{\pm h\}) = 0$. Therefore, (ii) is given from Lemma 15.15.

(i) To obtain (i), $b^n \longrightarrow b$ and $c_n \longrightarrow c$ are equivalent conditions when $\tilde{\nu}_n \xrightarrow{w} \tilde{\nu}$ and $\nu(\{\pm h\}) = 0$ as $|x - \frac{x}{1+x^2}| \leqslant |x|^3$.

□

Yet another result that will be used to prove the approximation theorem is the one below, inspired by Pollard's book.

Theorem : (Processes with independent increments)

Let $X, X_1, X_2, \cdots \in D[0,1]$, with all independent increments. Suppose X has continuous sample paths. Then $X_n \longrightarrow X$ if and only if :

(i) $X_n(0) \longrightarrow X(0)$

(ii) $\forall\ s < t \quad X_n(t) - X_n(s) \xrightarrow{d} X(t) - X(s)$

(iii) For $\delta > 0$, $\exists\ \alpha, \beta > 0$ and $n_0 \in \mathbb{N}$ such that $\forall\ n \geqslant n_0\ |t-s| < \alpha \implies \mathbb{P}(|X_n(t) - X_n(s)| \leqslant \delta) \geqslant \beta$

Remark : X is said to have **continuous sample paths** if, for $X = (X_t)_{t \geqslant 0}$, $X_\bullet(\omega) : \mathbb{R}_+ \longrightarrow \mathbb{R}$ is continuous (seeing $\mathbb{R}_+$ and $\mathbb{R}$ as topological spaces) for a.e. $\omega \in \Omega$.

Proof: (of the approximation theorem)
$\boxed{\Longleftarrow}$ Input $M_t^\varepsilon = \frac{L_t^\varepsilon}{\sigma(\varepsilon)}$, thus $\phi_{M_t^\varepsilon}(u) = \mathbb{E}(e^{iuM_t^\varepsilon}) = \exp\left(t\left(iub_\varepsilon + \int_{\mathbb{R}} (e^{iux} - 1 - iux \mathbb{1}_{\{|x|<1\}}) \nu_\varepsilon(dx)\right)\right)$
with $\nu_\varepsilon(B) = \nu(\sigma(\varepsilon) B \cap]-\varepsilon, \varepsilon[)$ for $B \in \mathcal{B}(\mathbb{R})$ and $b_\varepsilon = \frac{1}{\sigma(\varepsilon)} \int_{\min(\sigma(\varepsilon), \varepsilon) < |x| < \varepsilon} x\, \nu(dx)$.

M_t^ε has stationary indep. increments by the previous theorem; one can see that it suffices to show convergence at endpoints, *i.e.* : $M_1^\varepsilon \xrightarrow{d} W_1$.

And :

$$M^\varepsilon \xrightarrow{D} W \iff \begin{pmatrix} \int_{|x|<\kappa} x^2 \nu_\varepsilon(dx) \longrightarrow 1 \quad \forall\ \kappa > 0 \\ \nu_\varepsilon(\{|x| \geqslant 1\}) \longrightarrow 0 \\ b_\varepsilon \xrightarrow[\varepsilon \to 0]{} 0 \end{pmatrix}$$

By using Theorem on convergence of infinitely divisible distributions, notice that :

$$\int_{|x|<\kappa} \left(\frac{x}{\sigma(\epsilon)}\right)^2 \nu_\varepsilon(dx) = \frac{\sigma^2(\min(\sigma(\kappa\varepsilon), \varepsilon))}{\sigma^2(\varepsilon)}$$

So 9 is necessary for convergence in distribution.

$\boxed{\Longrightarrow}$ Observe that

$$\nu_\varepsilon(\{|x| \geqslant 1\}) = \nu(\{\min(\varepsilon, \sigma(\varepsilon)) \leqslant |x| \leqslant \varepsilon\}) \leqslant \frac{\sigma^2(\varepsilon) - \sigma^2(\min(\varepsilon, \sigma(\varepsilon)))}{\min(\varepsilon^2, \sigma^2(\varepsilon))} \longrightarrow 0$$

and :

$$|b_\varepsilon| \leqslant \frac{1}{\sigma^2(\varepsilon)} \int_{\min(\sigma(\varepsilon),\varepsilon)<|x|<\varepsilon} x^2 \nu(dx) \leqslant \frac{\sigma^2(\varepsilon) - \sigma^2(\min(\varepsilon, \sigma(\varepsilon)))}{\min(\varepsilon^2, \sigma^2(\varepsilon))} \longrightarrow 0$$

□

4.3.3 Convergence and error estimates

Proposition :

$$(9) \Longleftarrow \frac{\sigma(\epsilon)}{\epsilon} \xrightarrow{\epsilon \to 0} +\infty \tag{13}$$

The previous proposition highlights a relation between two conditions on volatility of the Brownian motion approximation, which can lead one to wonder if these two conditions are equivalent.

Example :

Let $(a_n)_n \quad a_n \longrightarrow 0$ decreasing with $a_1 = 1 \quad \frac{a_{n+1}}{a_n} \xrightarrow{n \to \infty} 0$.
By setting a Lévy measure ν such that $\sigma(\epsilon) = a_n \quad \forall \epsilon \in]a_{n+1}, a_n]$ and $\sigma(\epsilon) = 1 \quad \forall \epsilon > 1$.
As $\liminf_{\epsilon \to 0} \frac{\sigma(\epsilon)}{\epsilon} = 1$, (13) fails.
Conversely, $\forall \kappa > 0 \quad \exists n_0 \in \mathbb{N}$ such that $\forall n \geqslant n_0 \ \kappa a_n > a_{n+1}$ which implies that $\sigma(\min(\kappa\sigma(\epsilon), \epsilon)) = \sigma(\epsilon) \quad \forall \epsilon \in]0, a_{n_0}]$ so (9) is verified.

Examples :

- **Compound Poisson process**
  $\nu(\mathbb{R}) < +\infty$, $\sigma(\epsilon) = o(\epsilon)$. Therefore (9) is not verified.
- **Gamma process**
  $\nu(dx) = a\frac{e^{-bx}}{x}\mathbb{1}_{\{x>0\}}(x)\,dx$ for $a, b > 0$ with $\sigma(\epsilon) \sim \sqrt{\frac{a}{b}}\epsilon$ so normal approximation is not verified.
- **α-stable process** ($\alpha \in]0, 2[$)
  $\nu(dx) = \left(\frac{a}{|x|^{1+\alpha}}\mathbb{1}_{\{x<0\}} + \frac{b}{x^{1+\alpha}}\mathbb{1}_{\{x>0\}}\right)dx \quad a, b \geqslant 0,\ a + b > 0$, and $\sigma(\epsilon) = \sqrt{\frac{a+b}{2-\alpha}}\epsilon^{1-\frac{\alpha}{2}}$.
  Thus by the previous proposition the normal approximation is verified.
- **Normal inverse Gaussian process**
  $\sigma(\epsilon) \underset{\epsilon\to 0}{\sim} \sqrt{\frac{2\delta}{\pi}}\epsilon$ so $\frac{\sigma(\epsilon)}{\epsilon} \underset{\epsilon\to 0}{\sim} \sqrt{\frac{2\delta}{\pi\epsilon}} \xrightarrow{\epsilon\to 0} +\infty$ therefore here, again, the normal approximation is valid.

4.3.4 Discussion and limitations

Previously, we have listed a couple examples depicting the link between two conditions on characterizing elements of the approximating process to evaluate the nature of the convergence of the small jump part. This was the case of the two major results of the paper. Whereas other limits can appear if (9) is not verified.

Introduce $\left(\mathcal{L}\left(\frac{L_1^\epsilon}{\sigma(\epsilon)}\right)\right)_{0<\epsilon\leqslant 1}$, as $\mathbb{V}(L_1^\epsilon) = \sigma^2(\epsilon)$ the family is tight *i.e.* $\exists(\epsilon_n)_n$ decreasing and Z a Lévy process such that $\frac{L_1^{\epsilon_n}}{\sigma(\epsilon)} \xrightarrow{d} Z$. Looking at the convergence of a Lévy process is the same as looking at its convergence evaluated at time $t = 1$.
Then, consider :

$$I(\nu) = \bigcap_{\delta>0} \overline{\left\{\mathcal{L}\left(\frac{L_1^\epsilon}{\sigma(\epsilon)}\right) \mid 0 < \epsilon < \delta\right\}}$$

$I(\nu) \subset I_0 = \{X \text{ infinitely divisible } \mid \mathbb{E}(X) = 0,\ \mathbb{V}(X) \leqslant 1\}$
Remark : $\mathcal{L}$ stands for "convergence in distribution/law of".

Also, for $F \in I_0$, $F = F_1 * F_2$ where $\widehat{F}_1(u) = \mathbb{E}(e^{iuF_1}) = \exp\left(\int_{-\infty}^{+\infty}(e^{iux} - 1 - iux)\nu_2(dx)\right)$ and $F_2 \hookrightarrow \mathcal{N}(0, \sigma_2^2)$ with :

- $0 \leqslant \sigma_1^2 + \sigma_2^2 \leqslant 1$
- $\sigma_1^2 = \int_{-\infty}^{+\infty} x^2\nu_1(dx)$ defined similarly as in previously.

All this to lead us to the following result which concludes this section.

Theorem :

$\exists\nu$ such that $I(\nu) = I_0$ *i.e.* $\exists(L_t)_{t\geqslant 0}$, Lévy process such that for any Lévy process $(Z_t)_{t\geqslant 0}$ having $\mathcal{L}(Z_1) \in I_0$, $\exists(\epsilon_n)_n$ decreasing to 0 satisfying :

$$\frac{L^{\epsilon_n}}{\sigma(\epsilon)} \xrightarrow{d} Z$$

5 Lévy-Driven Stochastic Differential Equations

5.1 Stochastic Differential Equations with Jumps

5.1.1 General framework

Let $L = (L_t)_{t\geqslant 0}$ be a Lévy process being square-integrable. So $\exists\ b \in \mathbb{R}$, $c \in \mathbb{R}_+$ and ν a measure over $\mathbb{R}^*$ satisfying $\int_{\mathbb{R}^*} x^2\,\nu(dx) < \infty$ and L_t can take the form $\forall\, t \geqslant 0$:

$$L_t = bt + \sqrt{c}W_t + \int_0^t \int_{\mathbb{R}^*} x\,\tilde{N}(ds, dx)$$

as seen previously, ($(W_t)_{t\geqslant 0}$: standard Brownian motion, N: random Poisson measure with intensity $ds\,\nu(dx)$, $\tilde{N}$: associated compensated Poisson random measure).

The goal, now, is to solve the stochastic differential equation (SDE) given by :

$$X_t = x + \int_0^t \sigma(X_{s-})\,dL_s \quad \text{for } x \in \mathbb{R},\ \sigma : \mathbb{R} \to \mathbb{R} \tag{14}$$

Firstly, a numerical scheme seems to be a relevant approach for solving (14). Namely, an Euler scheme $(X^n_{\frac{i}{n}})_{n\geqslant 0}$ with time step $\frac{1}{n}$ is a starting idea. To do this, one can start by approximating increments of L by $\hat{\Delta}_i^{n,\epsilon} = L^\epsilon_{\frac{i}{n}} - L^\epsilon_{\frac{i-1}{n}}$ with L^ϵ being the same Lévy process compensated of its small jump term. It is also relevant to introduce $\Delta_i^{n,\epsilon} = \hat{\Delta}_i^{n,\epsilon} + U_i^{n,\epsilon}$ where $U_i^{n,\epsilon}$ is a Gaussian term with same expected value and variance as the small jump part. This is useful as it has been shown previously that approximating small jumps by Gaussian components is more efficient than neglecting them. After this, one denotes now $(\hat{X}^{n,\epsilon}_{\frac{\lfloor nt\rfloor}{n}})_{t\geqslant 0}$ (resp. $(X^{n,\epsilon}_{\frac{\lfloor nt\rfloor}{n}})_{t\geqslant 0}$) the associated Euler scheme combined with approximated increments $(\hat{\Delta}_i^{n,\epsilon})_{i\geqslant 1}$ (resp. $(\Delta_i^{n,\epsilon})_{i\geqslant 1}$).

5.2 Approximation Schemes for Lévy-Driven SDEs

5.2.1 Euler schemes with jumps

To pursue this purpose it is possible to set : $\forall\ \epsilon \in\ ]0,1[,\ k \in \mathbb{N}$

$$F_\epsilon(\nu) = \int_{|x|\geqslant\epsilon} \nu(dx) \quad m_k(\nu) = \int_{\mathbb{R}^*} |x|^k\,\nu(dx) \quad m_{k,\epsilon}(\nu) = \int_{|x|<\epsilon} |x|^k\,\nu(dx) \quad \beta_\epsilon(\nu) = \frac{m_{4,\epsilon}(\nu)}{m_{2,\epsilon}(\nu)}$$

Therefore : let $0 < \epsilon < 1,\ n \in \mathbb{N}$

$$\Delta_i^{n,\epsilon} = b_{n,\epsilon} + \sqrt{c}_{n,\epsilon}\,G + \sum_{i=1}^{N_{n,\epsilon}} Y_i^\epsilon \quad \text{with} \quad b_{n,\epsilon} = \frac{1}{n}\left(b - \int_{|x|\geqslant\epsilon} x\,\nu(dx)\right) \quad c_{n,\epsilon} = \frac{c + m_{2,\epsilon}(\nu)}{n} \tag{15}$$

- $G \sim \mathcal{N}(0,1)$
- $N_{n,\epsilon} \sim \mathcal{P}\left(\frac{F_\epsilon(\nu)}{n}\right)$
- $(Y_n^\epsilon)_{n\geqslant 1}$ i.i.d. sequence $Y_n^\epsilon \sim \frac{\mathbb{1}_{\{|x|\geqslant\epsilon\}}}{F_\epsilon(\nu)}\nu(dx) \quad \forall\ n \geqslant 1.$

So, naturally the following scheme appears :

$$X^{n,\epsilon}_{\frac{i+1}{n}} = X^{n,\epsilon}_{\frac{i}{n}} + \sigma(X^{n,\epsilon}_{\frac{i}{n}})\Delta^{n,\epsilon}_{i+1} \quad \forall\ i \geqslant 0 \quad \text{with } X_0^{n,\epsilon} = x \tag{16}$$

Remarks : (Miscellaneous)

- The simulation of $\Delta_i^{n,\epsilon}$ has a cost of order $1 + \mathbb{E}(N_{n,\epsilon}) = 1 + \frac{F_\epsilon(\nu)}{n}$ as the algorithm performs a fixed amount of work, this is included in "1", consisting in running the Gaussian generation, the variable update

as well as looking at occurrence of jumps. Also, we have seen previously that for "large" jumps (greater than ϵ), the component is accumulated as a compound Poisson process which is exactly the third member of the sum of the right hand side of (15).

Hence, the cost of a simulated path over time interval $[0,T]$, one has that $(X^{n,\epsilon}_{\rho_n(t)})_{t\in[0,T]}$ is of cost :

$$Tn \cdot (\text{cost}(\Delta^{n,\epsilon}_{i+1})) = Tn\left(1 + \frac{F_\epsilon(\nu)}{n}\right) = T(n + F_\epsilon(\nu)).$$

- $\Delta^{n,\epsilon}_{i+1}$ follows the same law as $L^\epsilon_{\frac{i+1}{n}} - L^\epsilon_{\frac{i}{n}} + U^{n,\epsilon}_i$. Note that the Gaussian term previously introduced thus has expectation value and variance of $\int_{\frac{i}{n}}^{\frac{i+1}{n}} \int_{|x|<\epsilon} x\,\tilde{N}(ds,dx)$ with $L^\epsilon_t = bt + \sqrt{c}W_t + \int_0^t \int_{|x|\geqslant\epsilon} x\,\tilde{N}(ds,dx)$.
- The main idea of the study is to find a satisfying enough coupling between $U^{n,\epsilon}_i$ and $\int_{\frac{i}{n}}^{\frac{i+1}{n}} \int_{|x|<\epsilon} x\,\tilde{N}(ds,dx)$.

So follows the main result of this idea :

5.3 Fournier (2013): Convergence Results

Theorem :

Let $\sigma : \mathbb{R} \to \mathbb{R}$ bounded and Lipschitz-continuous, $0 < \epsilon < 1$, $n \in \mathbb{N}$. There exists a coupling between a solution to (14) $(X_t)_{t\geqslant 0}$ and an approximated solution $(X^{n,\epsilon}_{\rho_n(t)})_{t\geqslant 0}$, in the sense of what is defined previously, such that :

$$\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X^{n,\epsilon}_{\rho_n(t)}|^2\right) \leqslant C_T\left(\frac{1}{n} + n\,\beta_\epsilon(\nu)\right)$$

where C_T depends only on T, b, c, σ and $m_2(\nu)$.

5.3.1 Main convergence theorems

To provide a proof to this result we first need to state a couple of subsidiary elements which will be put together to complete the demonstration. We will begin with three classical stochastic process results :

Theorem : (Doob maximal inequality)

Let $(X_n)_n$ be a submartingale, $\forall\ R > 0$:

$$\mathbb{P}(\max_{0\leqslant k\leqslant n}(X_k \geqslant R)) \leqslant \frac{1}{R}\mathbb{E}(X_n^+\,\mathbb{1}_{\{\max_{0\leqslant k\leqslant n}(X_k\geqslant R)\}}) \leqslant \frac{\mathbb{E}(X_n^+)}{R}$$

with $X^+ = \max(X, 0)$.

Corollary :

Let $(X_n)_n$ be a martingale. Then, $\forall\ p > 1$:

$$\mathbb{E}\left(\sup_{0\leqslant k\leqslant n} |X_k|^p\right) \leqslant \left(\frac{p}{p-1}\right)^p \mathbb{E}(|X_n|^p)$$

Additionally, note that for a Wiener process (*i.e.*, a Brownian motion) we have the following statement :
Let $(W_t)_{t\geqslant 0}$ be a Wiener process and $(X_t)_{t\geqslant 0}$ a stochastic process adapted to filtration $(\mathcal{F}_t)_{t\geqslant 0}$. Then the Itô isometry is verified, namely :

$$\mathbb{E}\left(\left(\int_0^t X_s\, dW_s\right)^2\right) = \mathbb{E}\left(\int_0^t X_s^2\, ds\right)$$

Lemma : (Grönwall Lemma)

If $\Psi \geqslant 0$ and φ are continuous functions satisfying :

$$\forall\ t \geqslant 0 \quad \varphi(t) \leqslant k + \int_0^t \Psi(s)\varphi(s)\, ds \quad k \in \mathbb{R}.$$

Then : $\forall\ t \geqslant 0$

$$\varphi(t) \leqslant k \exp\left(\int_0^t \Psi(s)\, ds\right)$$

Introducing the Euler scheme with step $\frac{1}{n}$ given by :

$$\Delta_i^n = L_{\frac{i}{n}} - L_{\frac{i-1}{n}}, \quad X_0^n = x, \quad X^n_{\frac{i+1}{n}} = X^n_{\frac{i}{n}} + \sigma(X^n_{\frac{i}{n}})\Delta^n_{i+1} \tag{17}$$

Proposition 1 :

For a Lévy process $(L_t)_{t\geqslant 0}$, we have, for $(X_t)_{t\geqslant 0}$ solution to (14) and for $(X^n_{\frac{i}{n}})_{i\geqslant 0}$ defined as above :

$$\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X^n_{\rho_n(t)}|^2\right) \leqslant \frac{C_T}{n}$$

where C_T only depends on $T, b, c, m_2(\nu)$ and σ.
We suppose σ to be Lipschitz continuous here.

Proof :
Consider :

$$\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X^n_{\rho_n(t)}|^2\right)$$

For $0 \leqslant s \leqslant v \leqslant T$
Then :

$$|X_v - X_s| = |x + \int_0^v \sigma(X_{u-})\, dL_u - x - \int_0^s \sigma(X_{u-})\, dL_u|$$

and :

$$\int_s^v \sigma(X_{u-})\, d(bu + \sqrt{c}W_u + \int_0^u \int_{\mathbb{R}*} z\, \tilde{N}(dv, dz)) = \int_s^v b\,\sigma(X_{u-})\, du + \int_s^v \sqrt{c}\,|\sigma(X_{u-})|\, dW_u + \int_s^v \sigma(X_{u-}) \int_{\mathbb{R}*} z\, \tilde{N}(du, dz)$$

as

$$d(\int_0^u \int_{\mathbb{R}*} z\, \tilde{N}(dv, dz)) = \int_{\mathbb{R}*} z\, \tilde{N}(du, dz)$$

so :

$$
\begin{aligned}
|X_v - X_s|^2 &\leqslant \left(\int_s^v |\sigma(X_{u-})|\, dL_u\right)^2 \\
&= \left(\int_s^v b\,|\sigma(X_{u-})|\, du + \int_s^v \sqrt{c}\,\sigma(X_{u-})\, dW_u + \int_s^v \sigma(X_{u-}) \int_{\mathbb{R}^*} z\, \tilde{N}(du, dz)\right)^2 \\
&\leqslant \left(b\int_s^v |\sigma(X_u)|\, du + \sqrt{c}\int_s^v \sigma(X_u)\, dW_u + \int_s^v \sigma(X_{u-}) \int_{\mathbb{R}^*} z\, \tilde{N}(du, dz)\right)^2 \\
&\leqslant 3\left(b\int_s^v |\sigma(X_u)|\, du\right)^2 + 3\left(\sqrt{c}\int_s^v \sigma(X_u)\, dW_u\right)^2 + 3\left(\int_s^v \int_{\mathbb{R}^*} \sigma(X_{u-}) z\, \tilde{N}(du, dz)\right)^2
\end{aligned}
$$

Notice that X_{u-} is approximated to X_u in the argument of function σ in the two first terms as X is càdlàg so has countable number of jumps which thus have measure 0 w.r.t. to the first integral and as $(W_t)_{t\geqslant 0}$ is a Brownian motion, it is a continuous martingale, adding the fact that jumps of X occur at random times, X_{u-} and X_u are equal a.e. w.r.t. the Brownian motion's path.
This cannot be done for the third as the integral would no longer be stochastic and predictable as jump measure $\tilde{N}$ has jumps taking place at the exact same time X might jump.
This can be seen when evaluating jumps of $(X_t)_{t\geqslant 0}$ with a previously mentioned object $\Delta X = (\Delta X_t)_{t\geqslant 0}$ being the jump process of $X = (X_t)_{t\geqslant 0}$. As this process is not naturally 0, we necessarily need to have jumps for the jump part due to the fact that $\Delta(bt) = \Delta(\sqrt{c}W_t)$.

So we have :

$$
\sup_{v\in[s,t]} |X_v - X_s|^2 \leqslant 3\left(b\int_s^t |\sigma(X_u)|\, du\right)^2 + \sup_{v\in[s,t]}\left(\sqrt{c}\int_s^v \sigma(X_u)\, dW_u\right)^2 + \sup_{v\in[s,t]}\left(\int_s^v \int_{\mathbb{R}^*} \sigma(X_{u-})\, z\, \tilde{N}(du, dz)\right)^2
$$

So :

$$
\begin{aligned}
&\mathbb{E}\left(\sup_{v\in[s,t]} |X_v - X_s|^2\right) \leqslant 3\,\mathbb{E}\left(\left(b\int_s^t |\sigma(X_w)|dw\right)^2 + \sup_{v\in[s,t]}\left(\sqrt{c}\int_s^v \sigma(X_w)dW_w\right)^2 + \sup_{v\in[s,t]}\left(\int_s^v \int_{\mathbb{R}^*} \sigma(X_{w-}) z\, \tilde{N}(dw, dz)\right)^2\right) \\
&= 3\left[\mathbb{E}\left(\left(b\int_s^t |\sigma(X_w)|dw\right)^2\right) + \mathbb{E}\left(\sup_{v\in[s,t]}\left(\sqrt{c}\int_s^v \sigma(X_w)dW_w\right)^2\right) + \mathbb{E}\left(\sup_{v\in[s,t]}\left(\int_s^v \int_{\mathbb{R}^*} \sigma(X_{w-}) z\, \tilde{N}(dw, dz)\right)^2\right)\right]
\end{aligned}
$$

by Cauchy-Schwarz and Doob inequalities

$$
\leqslant 3\,\mathbb{E}\left(\int_s^t b^2 dw \int_s^t |\sigma(X_w)|^2 dw\right) + 4c\,\mathbb{E}\left(\left(\int_s^t \sigma(X_w)dW_w\right)^2\right) + 4\,\mathbb{E}\left(\int_s^t \int_{\mathbb{R}^*} \sigma(X_{w-})^2 z^2 \nu(dz)dw\right)
$$

by Itô isometry as $(W_t)_{t\geqslant 0}$ is a Wiener process

$$
\begin{aligned}
&\leqslant 3\,\mathbb{E}\left(b^2(t-s)\int_s^t \|\sigma\|_\infty^2 dw\right) + 4c\,\mathbb{E}\left(\int_s^t (\sigma(X_w))^2 dw\right) + 4\,\mathbb{E}\left(\int_s^t \int_{\mathbb{R}^*} \sigma(X_{w-})^2 z^2 \nu(dz)dw\right) \\
&= 3\,\mathbb{E}\left(b^2(t-s)\|\sigma\|_\infty^2(t-s)\right) + 4c\,\mathbb{E}\left(\int_s^t \|\sigma\|_\infty^2 dw\right) + 4\,\mathbb{E}\left(\int_s^t \sigma(X_{w-})^2 m_2(\nu)dw\right) \\
&= 3\,\mathbb{E}\left(b^2\|\sigma\|_\infty^2(t-s)^2\right) + 4c\,\mathbb{E}\left(\|\sigma\|_\infty^2(t-s)\right) + 4\,\mathbb{E}\left(\int_s^t \sigma(X_{w-})^2 m_2(\nu)dw\right)
\end{aligned}
$$

as $T > t - s$

$$
\begin{aligned}
&= 3\,b^2\|\sigma\|_\infty^2(t-s)^2 + 4c\,\|\sigma\|_\infty^2(t-s) + 4\,\mathbb{E}\left(\int_s^t \|\sigma\|_\infty^2 m_2(\nu)dw\right) \\
&\leqslant 3\,b^2\|\sigma\|_\infty^2 T(t-s) + 4c\,\|\sigma\|_\infty^2(t-s) + 4\,m_2(\nu)\|\sigma\|_\infty^2(t-s) \\
&\leqslant C\int_s^t (b^2T + c + m_2(\nu))\|\sigma\|_\infty^2 dw = C_T(t-s)
\end{aligned}
$$

Furthermore, $X^n_{\rho_n(t)} = x + \int_0^{\rho_n(t)} \sigma(X^n_{\rho_n(s-)})\, dL_s$. Now setting $A^n_t = \mathbb{E}\left(\sup_{s\in[0,t]} |X_{\rho_n(s)} - X^n_{\rho_n(s)}|^2\right)$ so :

$$A^n_t = \sup_{s\in[0,t]} \mathbb{E}\left(\left|\int_0^{\rho_n(s)} (\sigma(X_{\rho_n(u-)}) - \sigma(X^n_{\rho_n(u-)}))\, dL_u\right|^2\right)$$

Therefore, similarly to the method above :

$$\begin{aligned}
\mathbb{E}(A^n_t) &\leqslant 3\,\mathbb{E}\left(\int_0^{\rho_n(t)} b^2\, du \int_0^{\rho_n(t)} |\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\, du\right) + 4c\,\mathbb{E}\left(\int_0^{\rho_n(t)} |\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\, du\right) \\
&+ 4\,\mathbb{E}\left(m_2(\nu)\int_0^{\rho_n(t)} |\sigma(X_{u-}) - \sigma(X^n_{\rho_n(u-)})|^2\, du\right) \\
&= C\left(b^2\,\rho_n(t)\,\mathbb{E}\left(\int_0^{\rho_n(t)} |\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\, du\right) + c\,\mathbb{E}\left(\int_0^{\rho_n(t)} |\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\, du\right)\right. \\
&\left.+ m_2(\nu)\,\mathbb{E}\left(\int_0^{\rho_n(t)} |\sigma(X_{u-}) - \sigma(X^n_{\rho_n(u-)})|^2\, du\right)\right. \\
&\leqslant C\left(b^2\,T\int_0^{\rho_n(t)} \mathbb{E}\left(|\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\right) du + c\int_0^{\rho_n(t)} \mathbb{E}\left(|\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\right) du\right. \\
&\left.+ m_2(\nu)\int_0^{\rho_n(t)} \mathbb{E}\left(|\sigma(X_{u-}) - \sigma(X^n_{\rho_n(u-)})|^2\right) du\right. \\
&= C\int_0^{\rho_n(t)} (b^2T + c + m_2(\nu))\mathbb{E}\left(|\sigma(X_u) - \sigma(X^n_{\rho_n(u)})|^2\right) du \\
&\leqslant C_T\int_0^t \mathbb{E}\left(|X_u - X_{\rho_n(u)} + X_{\rho_n(u)} - X^n_{\rho_n(u)}|^2\right) du \\
&\leqslant C_T\int_0^t \mathbb{E}\left(|X_u - X_{\rho_n(u)}|^2\right) + \mathbb{E}\left(|X_{\rho_n(u)} - X^n_{\rho_n(u)}|^2\right) du \\
&\leqslant C_T\int_0^t |u - \rho_n(u)| + \mathbb{E}\left(\sup_{s\in[0,u]} |X_{\rho_n(s)} - X^n_{\rho_n(s)}|^2\right) du = C_T\int_0^t (|u - \rho_n(u)| + \mathbb{E}(A^n_u))\, du
\end{aligned}$$

as $\mathbb{E}(|X_u - X_{\rho_n(u)}|^2) \leqslant |u - \rho_n(u)|$ by the same measure as previously.

So we have :

$$\begin{aligned}
\mathbb{E}(A^n_t) &\leqslant C_T\int_0^t |u - \rho_n(u)|\, du + C_T\int_0^t \mathbb{E}(A^n_u)\, du \quad \text{as } |u - \rho_n(u)| < \frac{1}{n} \\
&= \frac{C_T}{n} + C_T\int_0^t \mathbb{E}(A^n_u)\, du
\end{aligned}$$

Hence by Grönwall Lemma :

$$\mathbb{E}(A^n_t) \leqslant \frac{C_T}{n} e^{C_T t} \quad \text{hence up to a constant } C_T,$$

$$\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X^n_{\rho_n(t)}|^2\right) \leqslant \frac{C_T}{n}$$

□

For the sake of length, we will not detail the proof of the next results but all of them follow the same kind of thought process.

Lemma 2 :

Let $n \in \mathbb{N}$ and $\epsilon > 0$. $\exists\ (\Delta_i^n)_{i\geqslant 1}, (\Delta_i^{n,\epsilon})_{i\geqslant 1}$ coupled families of i.i.d. random variables distributed respectively as in (17) and (15) such that $\forall\ i \geqslant 1$:

$$\mathbb{E}(|\Delta_i^n - \Delta_i^{n,\epsilon}|) \leqslant C\,\beta_\epsilon(\nu)$$

having C a universal constant. In addition, $\forall\ \epsilon > 0, \forall\ n \in \mathbb{N}, \forall\ i \geqslant 1$

$$\mathbb{E}(\Delta_i^n) = \mathbb{E}(\Delta_i^{n,\epsilon}) = \frac{b}{n} \quad \mathbb{V}(\Delta_i^n) = \mathbb{V}(\Delta_i^{n,\epsilon}) = \frac{c + m_2(\nu)}{n}$$

In addition, consider :

$$Y_i^{n,\epsilon} = X_{i/n}^n - X_{i/n}^{n,\epsilon}$$

Lemma 3 :

$\exists\ C$, constant depending on $b, c, m_2(\nu)$ and σ such that $\forall\ N \geqslant 1$:

$$\mathbb{E}\left(\sup_{i\in[\![0,N]\!]} |Y_i^{n,\epsilon}|^2\right) \leqslant Cn\,\beta_\epsilon(\nu)\left(1 + \frac{C}{n}\right)^n \left(1 + \left(\frac{N}{n}\right)^2\right)$$

Lemma 4 :

Consider $(\Delta_i^n, \Delta_i^{n,\epsilon})$ defined in *Lemma 2* and then $(X_{\frac{i}{n}}^n)_{i\geqslant 0}$ and $(X_{\frac{i}{n}}^{n,\epsilon})_{i\geqslant 0}$ defined in (17) and (16) as $\forall\ T \geqslant 0$:

$$\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)}^n - X_{\rho_n(t)}^{n,\epsilon}|^2\right) \leqslant C_T\, n\,\beta_\epsilon(\nu)$$

where C_T depends solely on $b, c, m_2(\nu)$ and σ.

Combining the elements we worked our way towards :

Proof : (of the main theorem)
Let $n \in \mathbb{N}$ and $\epsilon > 0$. Denote $Q(du, dv)$ the joint law of $(\Delta_1^n, \Delta_1^{n,\epsilon})$ constructed in *Lemma 2*.
Write $Q(du, dv) = Q_x(du)R(u, dv)$ with $Q_x(du)$ law of Δ_1^n and $R(u, dv)$ law of $\Delta_1^{n,\epsilon}|\Delta_1^n = u$.
Consider a Lévy process $(L_t)_{t\geqslant 0}$ as defined in the beginning of the section with the corresponding $(X_t)_{t\geqslant 0}$ solution to (14).
$\forall\ i \geqslant 0$ set : $\Delta_i^n = L_{\frac{i}{n}} - L_{\frac{i-1}{n}}$ and consider the Euler scheme $(X_{\frac{i}{n}}^n)_{i\geqslant 0}$ as in (17).
$\forall\ i \geqslant 1$, let $\Delta_i^{n,\epsilon}$ be distributed according to $R(\Delta_i^n, dv)$ in such a way that $(\Delta_i^{n,\epsilon})_{i\geqslant 1}$ is an i.i.d. sequence.
Moreover, let $(X_{\frac{i}{n}}^{n,\epsilon})_{t\geqslant 0}$ as in (16).
This technique gives that processes $(X_t)_{t\geqslant 0}, (X_{\frac{i}{n}}^n)_{i\geqslant 0}$ and $(X_{\frac{i}{n}}^{n,\epsilon})_{i\geqslant 0}$ are coupled, thus making sense to apply *Proposition 1* and *Lemma 4*. One gets :

$$\begin{aligned}\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X_{\rho_n(t)}^{n,\epsilon}|^2\right) &\leqslant 2\,\mathbb{E}\left(\sup_{t\in[0,T]} |X_{\rho_n(t)} - X_{\rho_n(t)}^n|^2 + \sup_{t\in[0,T]} |X_{\rho_n(t)}^n - X_{\rho_n(t)}^{n,\epsilon}|^2\right) \\ &\leqslant C_T\left(\frac{1}{n} + n\,\beta_\epsilon(\nu)\right)\end{aligned}$$

□

5.3.2 Generalization of the main result

By the previously done work, we can attempt to generalize the study for what we had previously defined.
A slight generalization of the latter main theorem comes when introducing :

$$L_t = bt + \sqrt{c}W_t + \int_0^t \int_{|z|<1} z\tilde{N}(ds,dz) + \int_0^t \int_{|z|\geqslant 1} zN(ds,dz)$$

being nothing other than the famous Lévy-Itô decomposition with standard Brownian motion $(W_t)_{t\geqslant 0}$ independent of $N(ds,dz)$ Poisson random measure on $\mathbb{R}_+ \times \mathbb{R}$ with intensity $ds\,\nu(dz)$ and $\tilde{N}(ds,dz)$ compensated Poisson measure.
Suppose that σ is locally Lipschitz-continuous with at most linear growth so the SDE admits a unique càdlàg strong solution $(X_t)_{t\geqslant 0}$.
Consider similarly to (15) $(\Delta_i^{n,\varepsilon})_{i\geqslant 1}$ sequence of random i.i.d. random variables given by, for $n \geqslant 1, 0 < \varepsilon < 1$:

$$\Delta_i^{n,\varepsilon} = b_{n,\varepsilon} + \sqrt{c}_{n,\varepsilon} G + \sum_{i=1}^{N_{n,\varepsilon}} Y_i^{\varepsilon}$$

with $b_{n,\varepsilon} = \frac{1}{n}\left(b - \int_{\varepsilon<|z|<1} z\,\nu(dz)\right)$, $c_{n,\varepsilon} = \frac{1}{n}(c + m_{2,\varepsilon}(\nu))$

- $G \sim \mathcal{N}(0,1)$
- $N_{n,\varepsilon} \sim \mathcal{P}(\frac{F_\varepsilon(\nu)}{n})$
- $(Y_i^{\varepsilon})_{i\geqslant 1}$ i.i.d. sequence with $Y_n \sim \frac{\mathbb{1}_{\{|z|>\varepsilon\}}\nu(dz)}{F_\varepsilon(\nu)} \quad \forall n \geqslant 1$

The only difference with what was defined previously is the value of $b_{n,\varepsilon}$. So naturally yields the result solving (14) :

Theorem :

Suppose σ is locally Lipschitz-continuous with at most linear growth and that $\int_{\mathbb{R}} \min(z^2,1)\nu(dz) < +\infty$. Then, $\forall n \in \mathbb{N}, 0 < \varepsilon < 1, \exists (X_t)_{t\geqslant 0}$ and $(X^{n,\varepsilon}_{\rho_n(t)})_{t\geqslant 0}$ coupling between solution to (14) and approximate solution such that $\forall\ T > 0$:

$$\lim_{A\to+\infty} \sup_{\substack{n\in\mathbb{N}\\ 0<\varepsilon<1}} \mathbb{P}\left(\frac{1}{\sqrt{\frac{1}{n} + n\,\beta_\varepsilon(\nu)}} \sup_{t\in[0,T]} |X_{\rho_n(t)} - X^{n,\varepsilon}_{\rho_n(t)}|^2 \geqslant A\right) = 0$$

5.3.3 Illustration for numerical simulation

Finally, to conclude our study we consider the case of a previously mentioned type of Lévy process: the stable process.
Consider the Lévy process given by :

$$L_t = \int_0^t \int_{|z|<1} z\,\tilde{N}(ds,dz) + \int_0^t \int_{|z|\geqslant 1} z\,N(ds,dz)$$

with $n(dz) = \frac{1}{|z|^{\alpha+1}}\,dz$.
Fixing, for example, $\alpha = 1.5$ and considering again (14) starting at 0 w.r.t. the latter Lévy process :

$$X_t = \int_0^t \sigma(X_{s-})\,dL_s$$

with $\sigma(y) = \frac{1+y^2}{1+y^4}$, one can, thus, set $V = \sup_{t\in[0,1]}(X_t)$.

Therefore, one also introduces for the last time the exact Euler scheme $(X^n_{\frac{i}{n}})_{i\geqslant 0}$ with time step $\frac{1}{n}$.
Let $\varepsilon > 0$, reconsider scheme $(X^{n,\varepsilon}_{\frac{i}{n}})_{i\geqslant 0}$ defined in (15) and (16) as $(X^{n,\varepsilon}_{\frac{i}{n}})_{i\geqslant 0}$ scheme where jumps smaller than ε are neglected.
Setting finally : $V^n = \sup_{i\in[\![0,n]\!]}(X^n_{\frac{i}{n}})$, $V^{n,\varepsilon} = \sup_{i\in[\![0,n]\!]}(X^{n,\varepsilon}_{\frac{i}{n}})$, $\hat{V}^{n,\varepsilon} = \sup_{i\in[\![0,n]\!]}(\hat{X}^{n,\varepsilon}_{\frac{i}{n}})$.
A couple of results of, that for values of $n \geqslant 1000$, simulations for $V^{2000,\varepsilon}$ and $\hat{V}^{2000,\varepsilon}$ compared to V^{2000} show conclusive results; being that the law of $\hat{V}^{2000,\varepsilon}$ is much closer to the law of V^{2000} than $V^{2000,\varepsilon}$ is, and for an insignificantly greater time of simulation. This finally leads us to draw the conclusion that approximating small jumps by Gaussian random variables is a more optimal method than simply neglecting them, which confirms the exact idea depicted in this whole paper.

Conclusion

This dissertation has navigated the intersection of abstract probability theory and practical numerical analysis, focusing on the robust simulation of Lévy processes. We began with the foundational premise that while the jump-diffusion framework offers unparalleled flexibility in modeling discontinuous phenomena, the presence of infinite activity near the origin presents a significant bottleneck for traditional simulation techniques. Through a systematic exploration of the **Lévy-Itô decomposition**, we have demonstrated that the complexity of a process is intrinsically tied to the behavior of its Lévy measure ν.
The core contribution of this research centers on the validation of the **Asmussen-Rosiński approximation**. Our findings confirm that the discarded energy of small jumps—those smaller than a chosen threshold ϵ—is not merely "noise" that can be ignored. Rather, by substituting these jumps with a Brownian motion whose variance is calibrated to the truncated measure, we significantly enhance the fidelity of the simulation. This approach effectively transforms a computationally impossible task into a highly efficient Gaussian-augmented jump-diffusion model.
Key insights derived from this study include :

1. **Theoretical Robustness**: The use of the (b, c, ν) triplet allows for a modular simulation architecture. Large jumps can be handled via Poissonian sampling, while the cumulative effect of small jumps is smoothly integrated into the diffusion term.

2. **Numerical Stability**: Our analysis of **Lévy-driven SDEs** shows that the Gaussian approximation maintains strong convergence properties. Following the results established in modern literature, we have shown that the error introduced by truncation vanishes at a predictable rate as $\epsilon \to 0$, provided the time-step of the Euler scheme is appropriately synchronized.

3. **Practical Application**: For practitioners in quantitative finance and physics, these methods provide a vital toolkit for pricing path-dependent options or modeling turbulent flows where the "fine structure" of the process is critical to the final outcome.

While this work—heavily informed by the seminal texts of **Sato (1999)** and **Applebaum (2009)**—establishes a comprehensive framework, it also opens several avenues for future inquiry. One promising direction is the extension of Gaussian substitution to **multivariate Lévy processes**, where the dependence structure between jumps must be preserved. Additionally, the development of **adaptive truncation** methods, where ϵ is updated dynamically based on the local curvature of the SDE's coefficients, could further optimize the balance between speed and precision.
Ultimately, this dissertation reaffirms that the gap between infinite mathematical structures and finite computational reality can be bridged through elegant approximation, ensuring that the power of Lévy processes remains accessible for real-world modeling.

Appendix

Numerical illustrastion

For the study of the last document, here, we illustrate the famously mentioned Gaussian approximation precision by comparing it to solution using Euler schemes either neglecting or approximation small jumps :

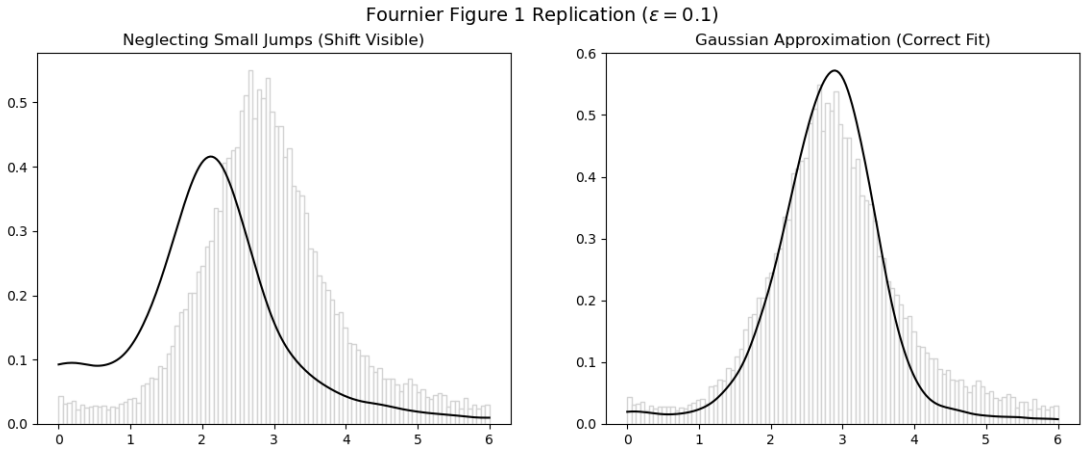


Figure 1: Simulation of solutions either neglecting or approximating small jumps.

The complexity is approximately $O(n \times M)$, where n is the number of time steps and M is the number of simulated paths.
The core idea in the Asmussen and Rosinski (2001) paper is that when simulating a Lévy process with infinite activity, you can approximate the "small jumps" (those smaller than some ϵ) as a Brownian motion.

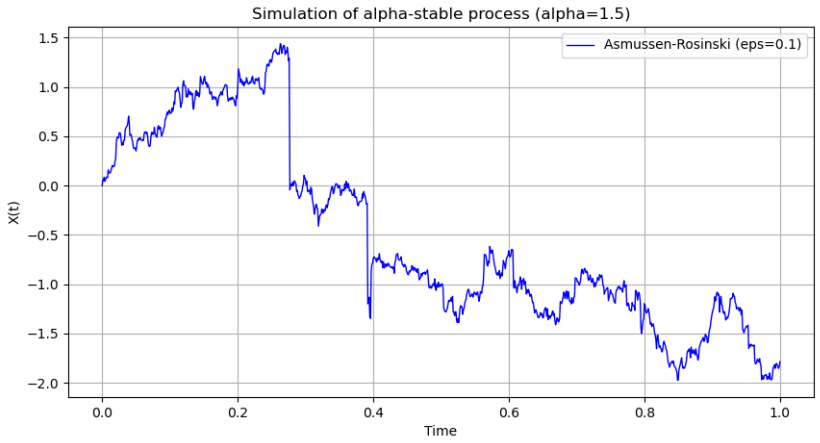


Figure 2: Simulation of a standard α-stable process ($1 < \alpha < 2$) using the Asmussen-Rosinski approximation.

The computational cost for the Asmussen-Rosinski method is expressed as $O(n + \epsilon^{-\alpha})$, where n represents the fixed cost of the Gaussian grid and $\epsilon^{-\alpha}$ represents the jump intensity that increases as the threshold shrinks.
All Python scripts used for the simulations in this dissertation are available at :

https://github.com/shelovesmoussey/levy-simulations

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